ASTR 2:
Galaxies and the Universe

Lecture #15 - Spring 2025

Prof. Ethan Nadler (he/him)

The Antennae Galaxy
Image Credit: Subaru, NAOJ, NASA/ESA/Hubble, R.W. Olsen

Announcements

Now is a good time to look at your grades and see what you need to do on the remaining assignments and final exam to get the grade you want in the class.

Calculate average HW, RQ percentage dropping lowest 2 scores.

$$\text{Total \%} = 0.25\times\textbf{HW\%} + 0.15\times\textbf{RQ\%} + 0.15\times\textbf{MT1\%} + 0.15\times\textbf{MT2\%} + 0.3\times\textbf{Final\%}$$

try out various numbers to see how you need to do

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Today's Guest Lecturer

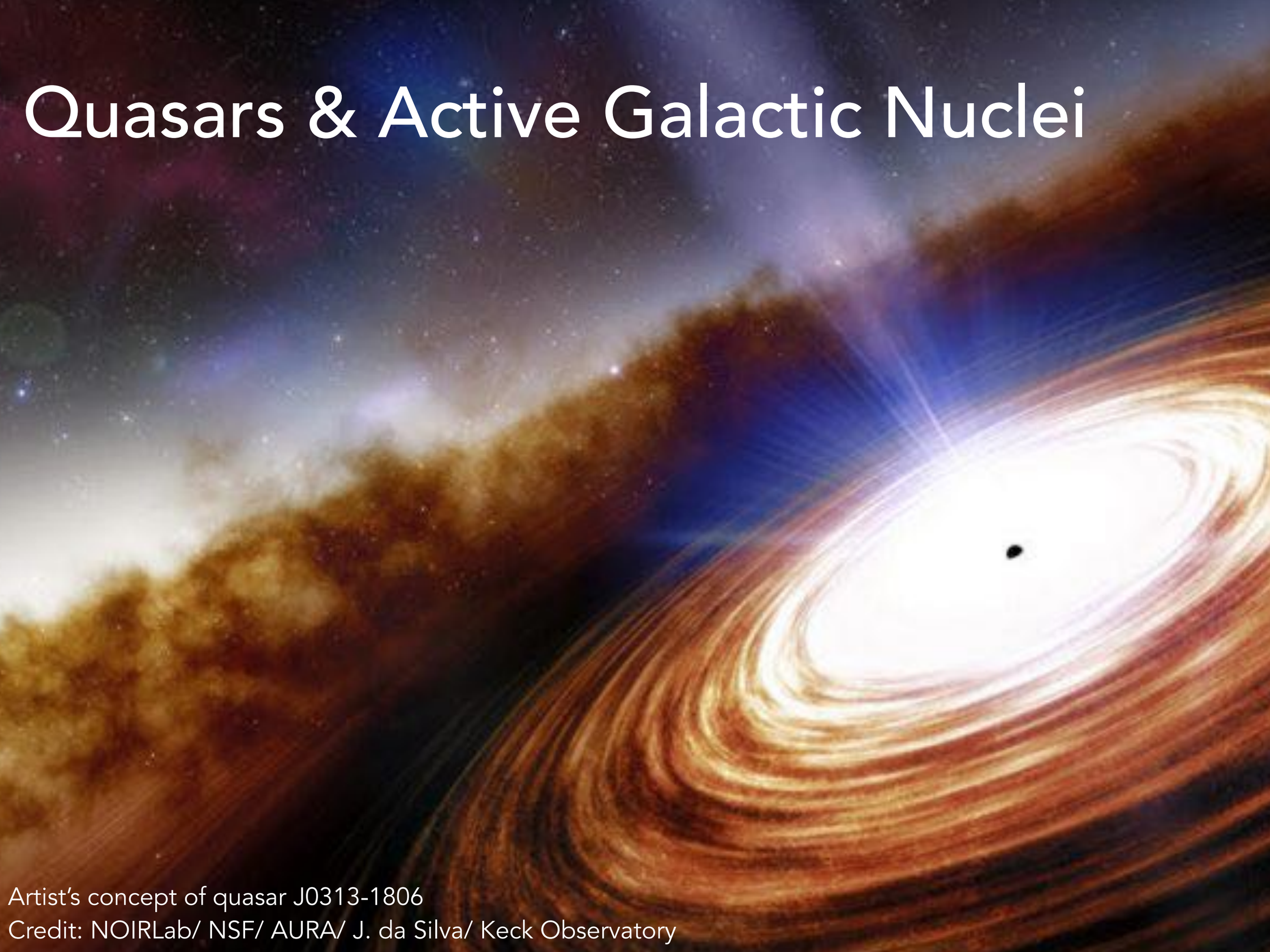
- Dr. Devontae Baxter is an NSF postdoctoral fellow in the Astronomy & Astrophysics department at UCSD.

- His research centers on using multi-wavelength observations & cosmological simulations to **investigate how large-scale galactic environment drive galaxy formation and evolution.**



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Quasars & Active Galactic Nuclei



Artist's concept of quasar J0313-1806
Credit: NOIRLab/ NSF/ AURA/ J. da Silva/ Keck Observatory

Quasars

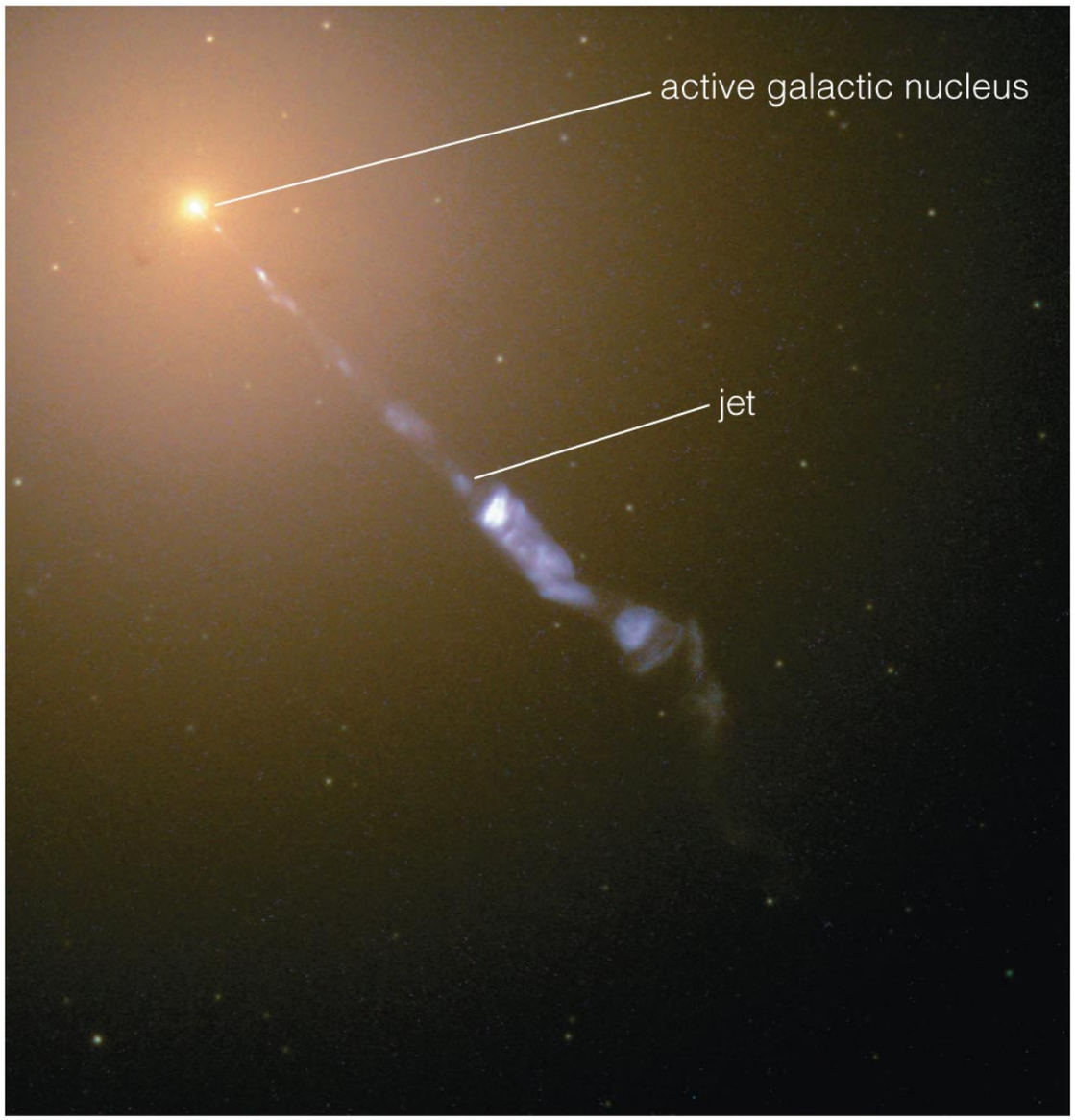
Quasars are very distant (among the most distant objects known) and were more common in the past - today they are relatively rare.

They are extremely luminous - some are over 10^{12} times brighter than the Sun!

Their brightness can change quickly - they vary on short timescales.

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Active Galactic Nuclei



Quasars are the *brightest* type of Active Galactic Nuclei (AGN).

These are galaxies with very bright centers - a lot of light is coming from a small region.

Many seem to produce jets of material.

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Survey Question

Which creates more energy, fusion in a star or material falling into a supermassive black hole?

A) Fusion in a star.

B) Material falling into a supermassive black hole.

C) They produce similar amounts of energy.

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Survey Question

Where does the energy released by a supermassive black hole come from?

A) Multiple supernovae.

B) Gravity.

C) Colliding stars near the black hole.

D) Nuclear energy.

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Energy from material falling towards a black hole.

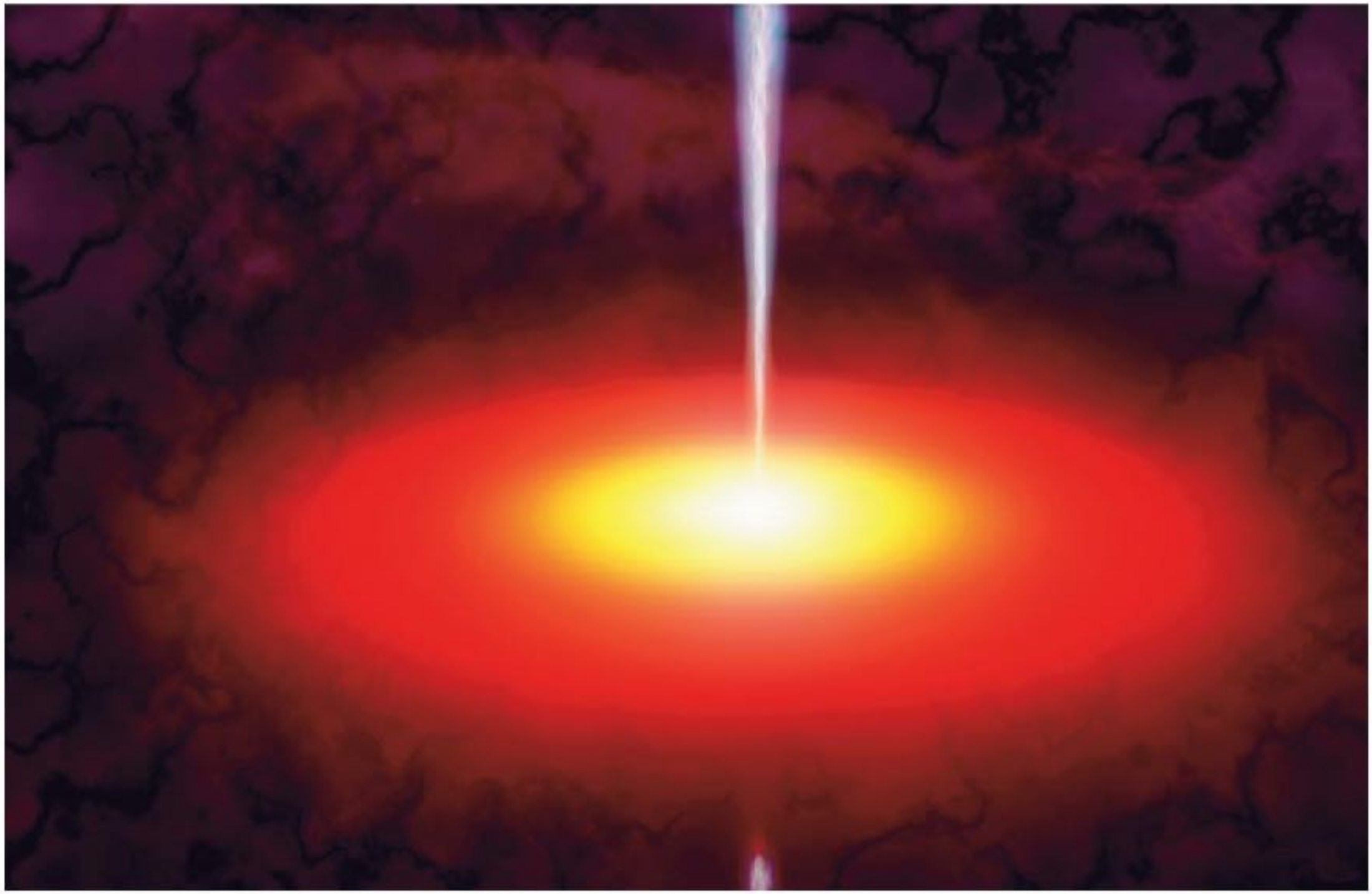
- Gravitational potential energy of matter falling towards black hole turns into kinetic energy.

- Friction in accretion disk turns kinetic energy into thermal energy (heat).

- Heat produces thermal radiation (photons).

- This process can convert 10-40% of $E = mc^2$ into radiation!

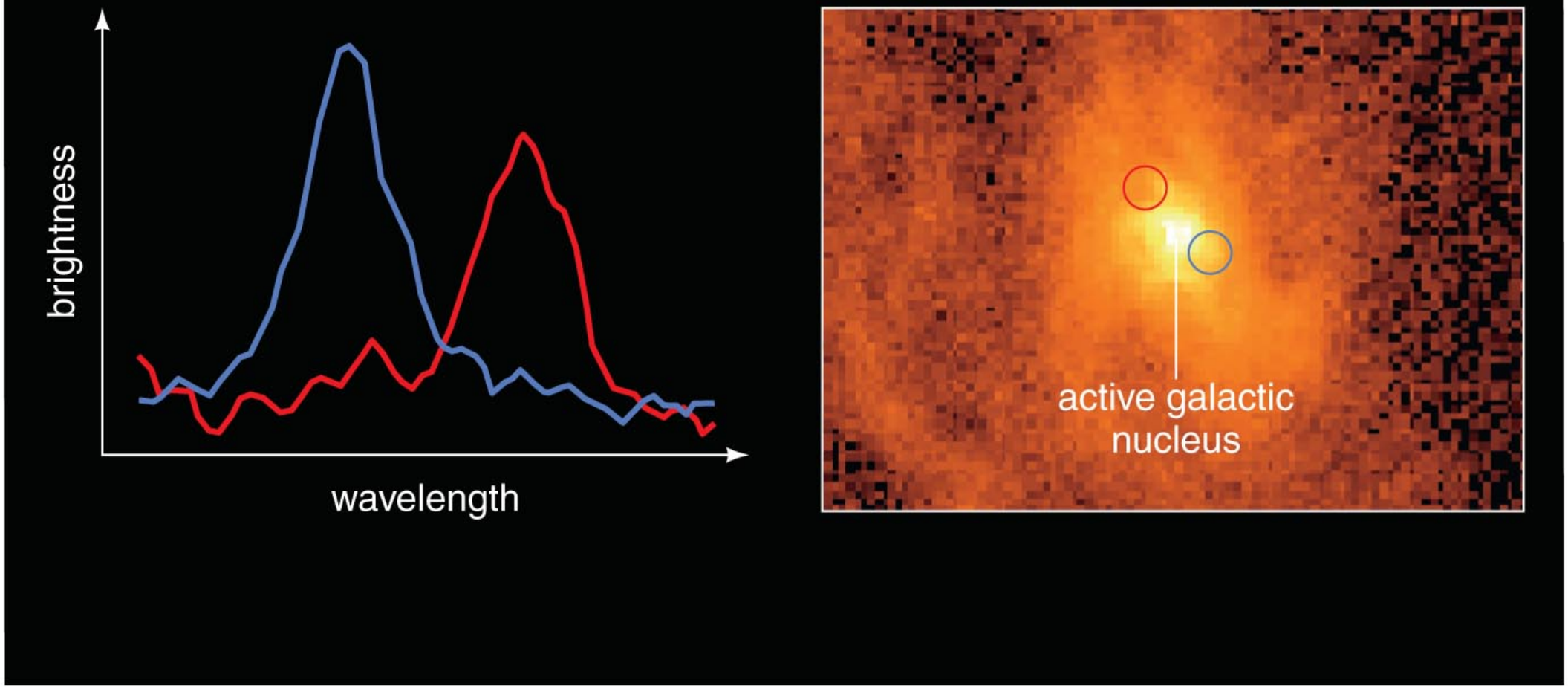
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Accretion of gas onto a supermassive black hole appears to be the only way to explain all the properties of quasars.

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Nuclear region of M87



Can get mass from velocity and radius: M87 black hole has 2.7 Billion solar masses!

Almost 1000 times more massive than the central supermassive black hole in the Milky Way

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M87 Black Hole



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Supermassive Black Holes and their Host Galaxies

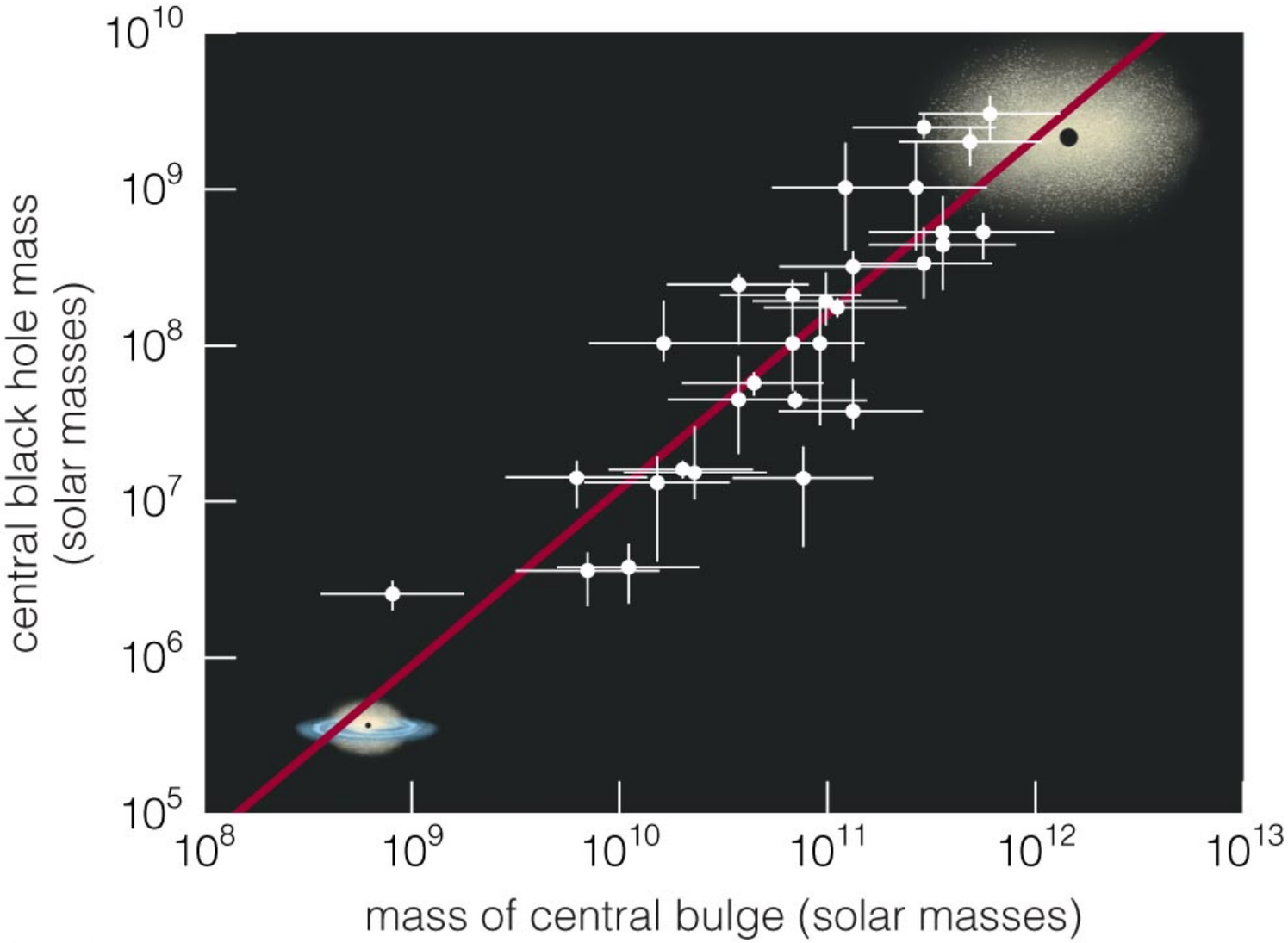
Almost every galaxy that we can see in enough detail has a supermassive black hole at its center!

They have a wide range of masses, just as their host galaxies do.

<u>Supermassive black holes are very common!</u>

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Connection between Galaxy Evolution and Supermassive BHs



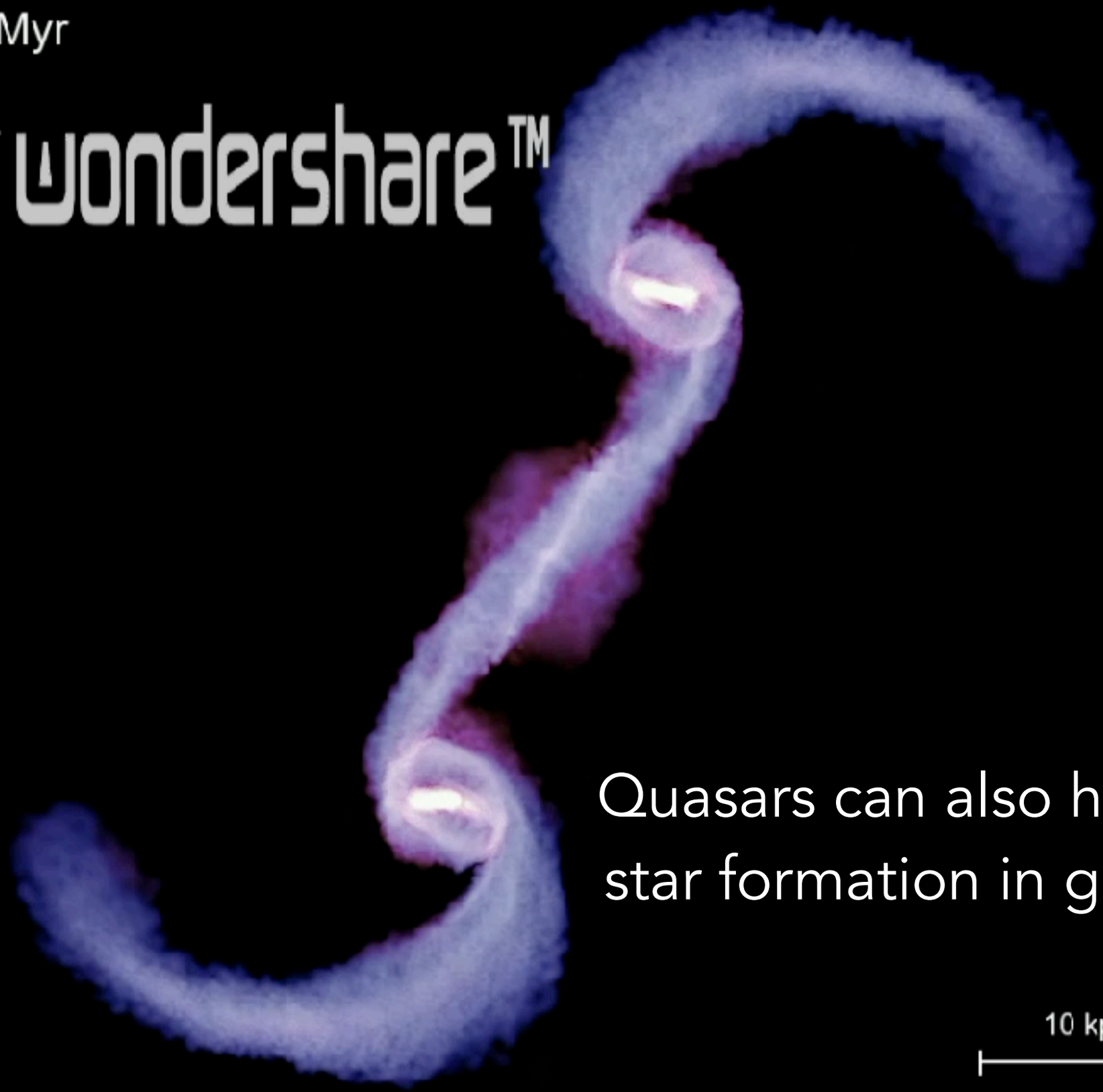
Correlation between black hole mass and stellar *bulge* mass

More recently, people have also found a correlation with total stellar mass.

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T = 600 Myr

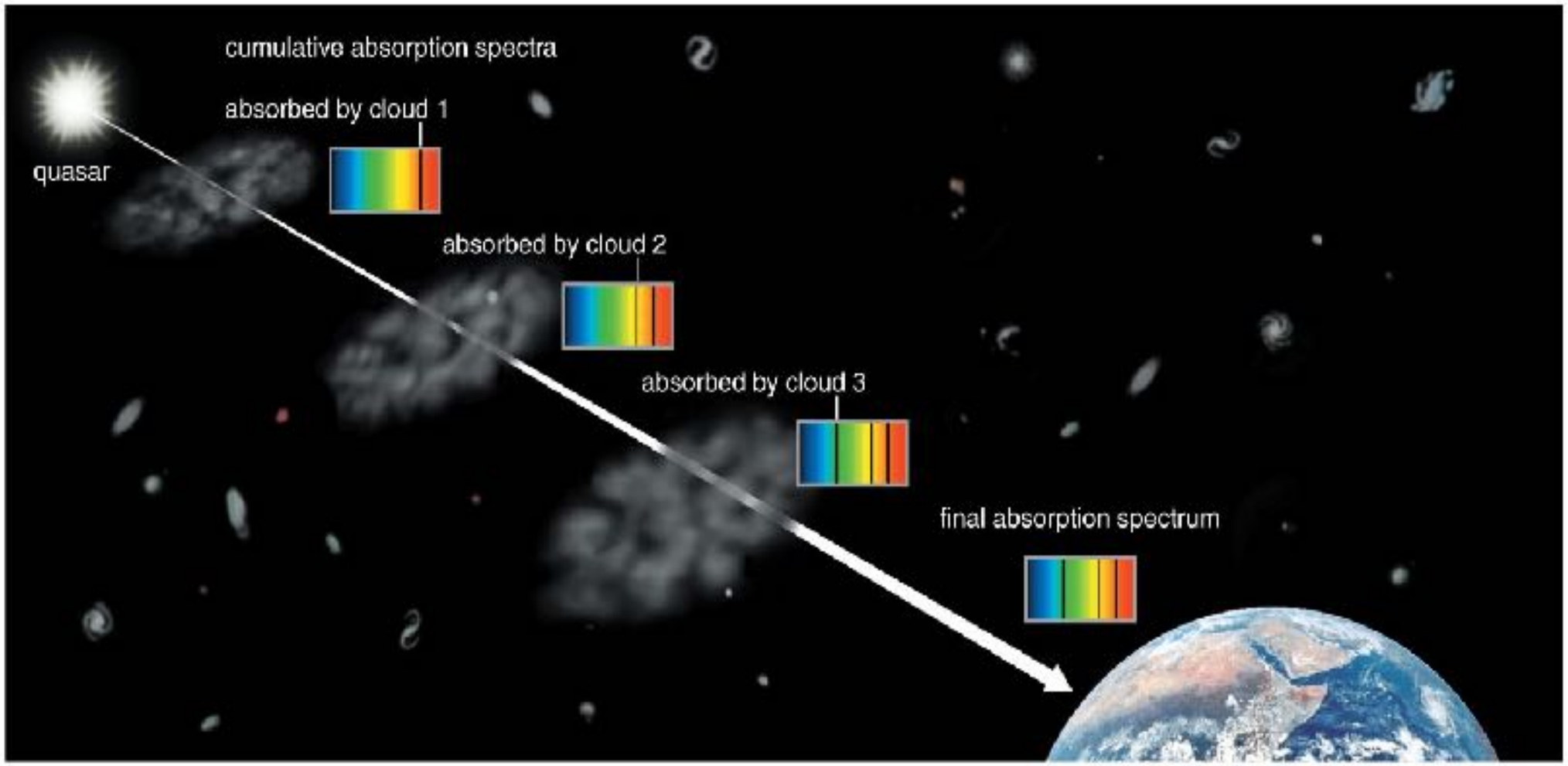
wondershare™



Quasars can also help halt star formation in galaxies.

10 kpc/h

Quasars as "Flashlights"



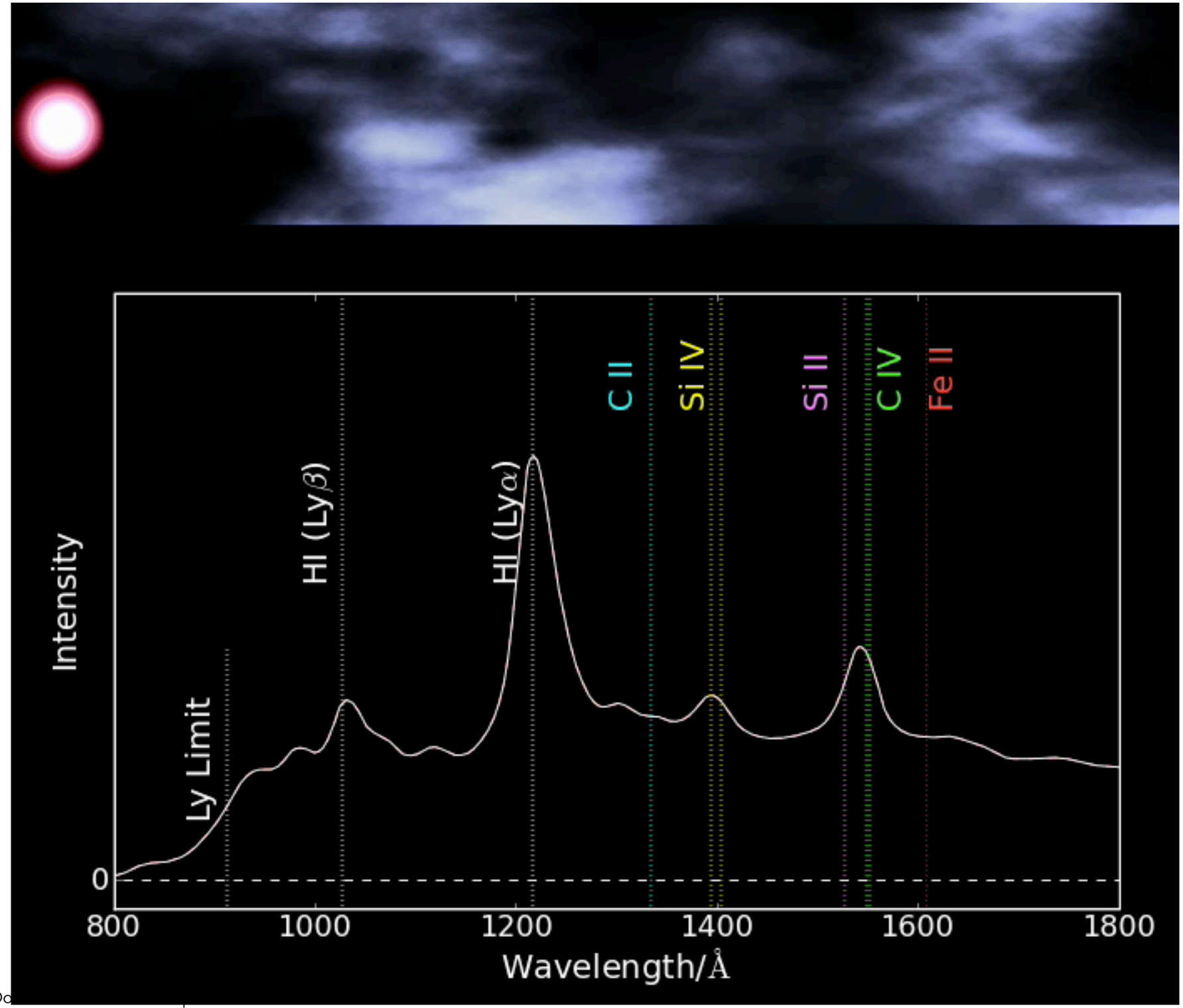
© 2010 Pearson Education, Inc.

Most quasars are very far away. Gas clouds between a quasar and Earth absorb some of the quasar's light.

We can learn about protogalactic clouds by studying the absorption lines they produce in quasar spectra.

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Quasars as "Flashlights"

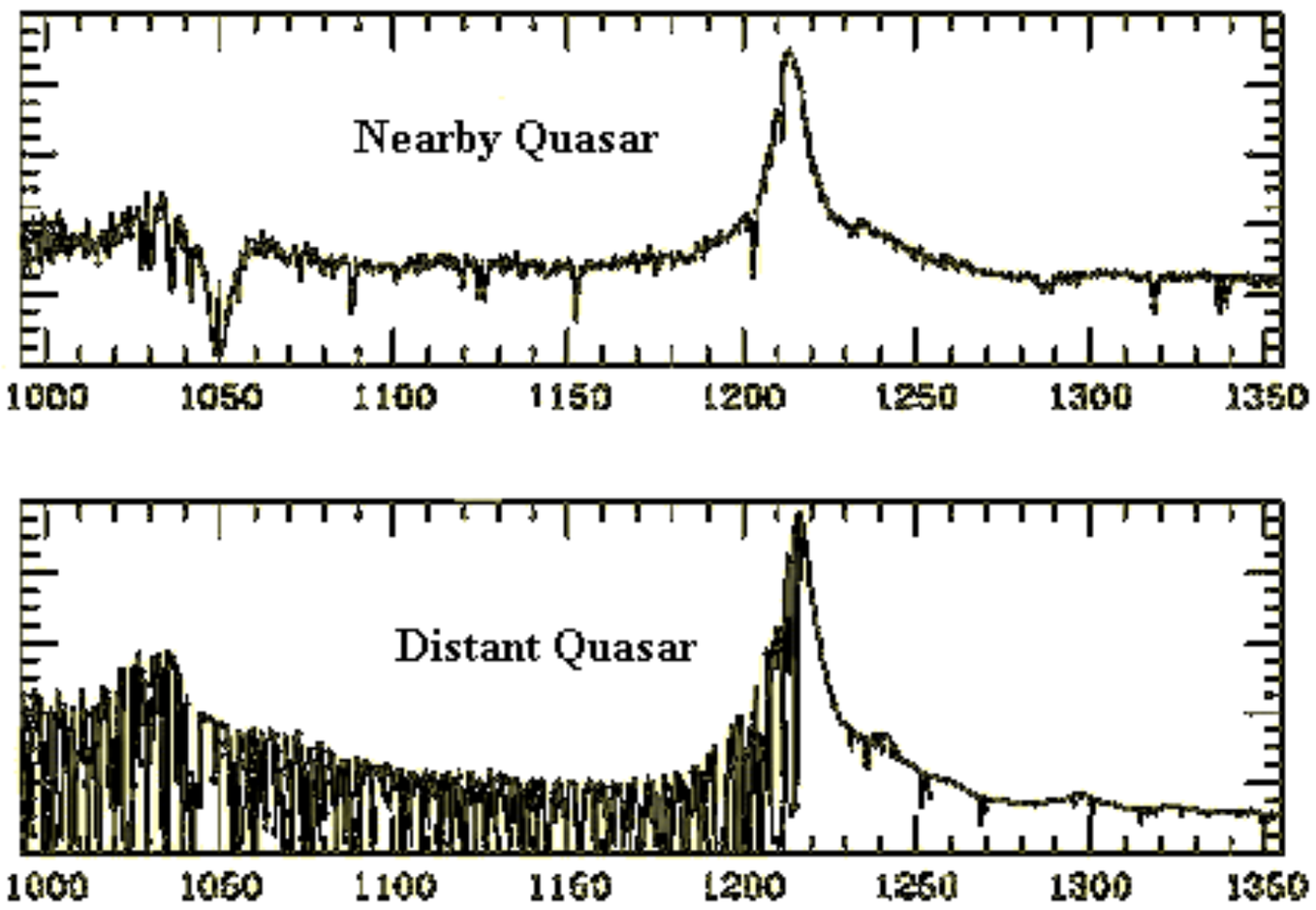


http://www.cosmocrunch.co.uk/?page=talk

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Quasars as "Flashlights"

See the gas between Earth and the quasar in absorption!



Very hard to see intergalactic gas in emission - diffuse, cold, doesn't radiate much.

Can see absorption at different wavelengths, depending on where the gas is in the expanding universe.

We have learned a great deal about distant gaseous clouds using quasars.

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Survey Question

How do we learn about protogalactic clouds?

A) by looking at emission lines from molecules in the clouds

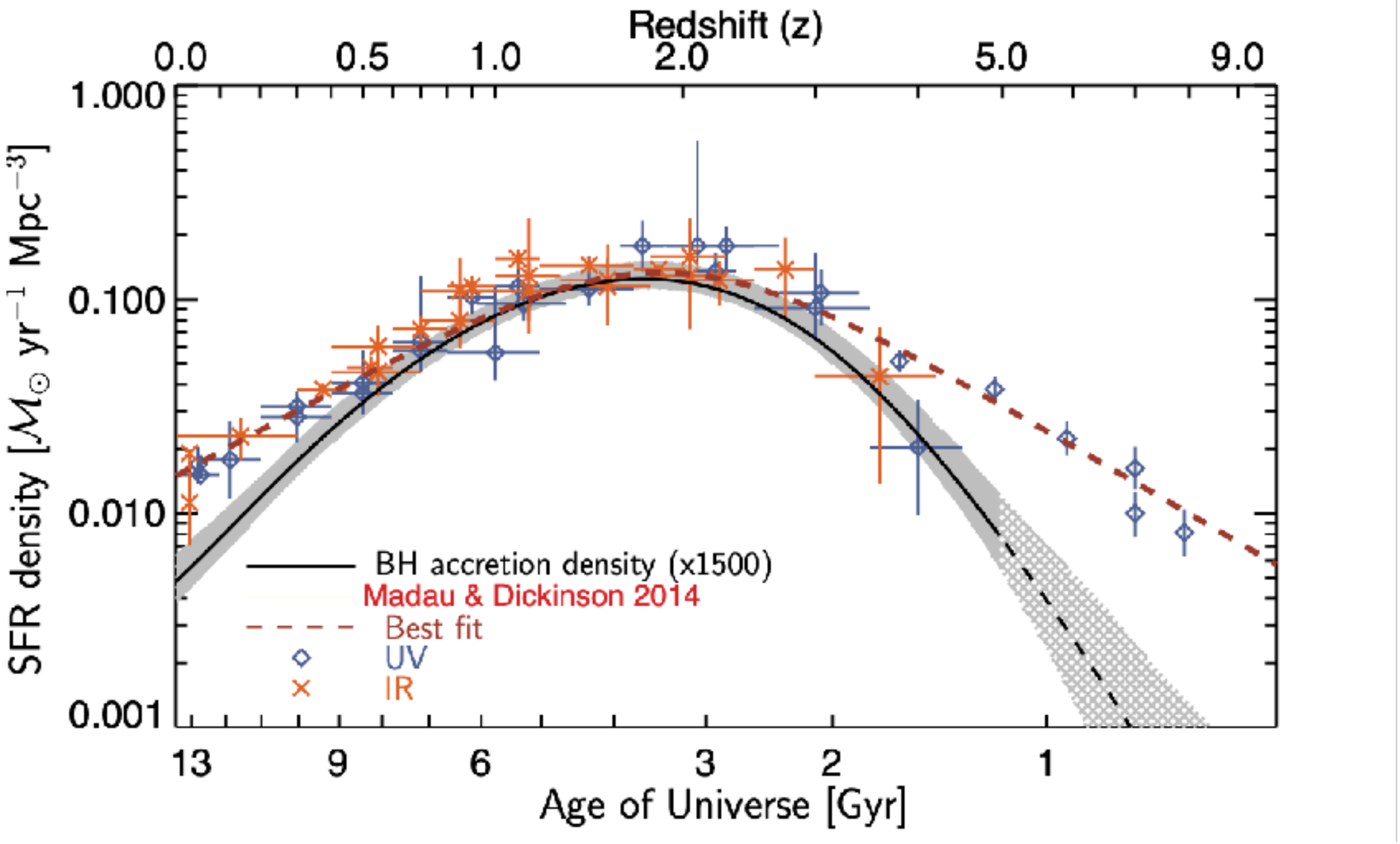
B) by looking at emission lines from hydrogen in the clouds

C) by looking at absorption line in spectra of background quasars.

D) by looking at thermal emission from the clouds.

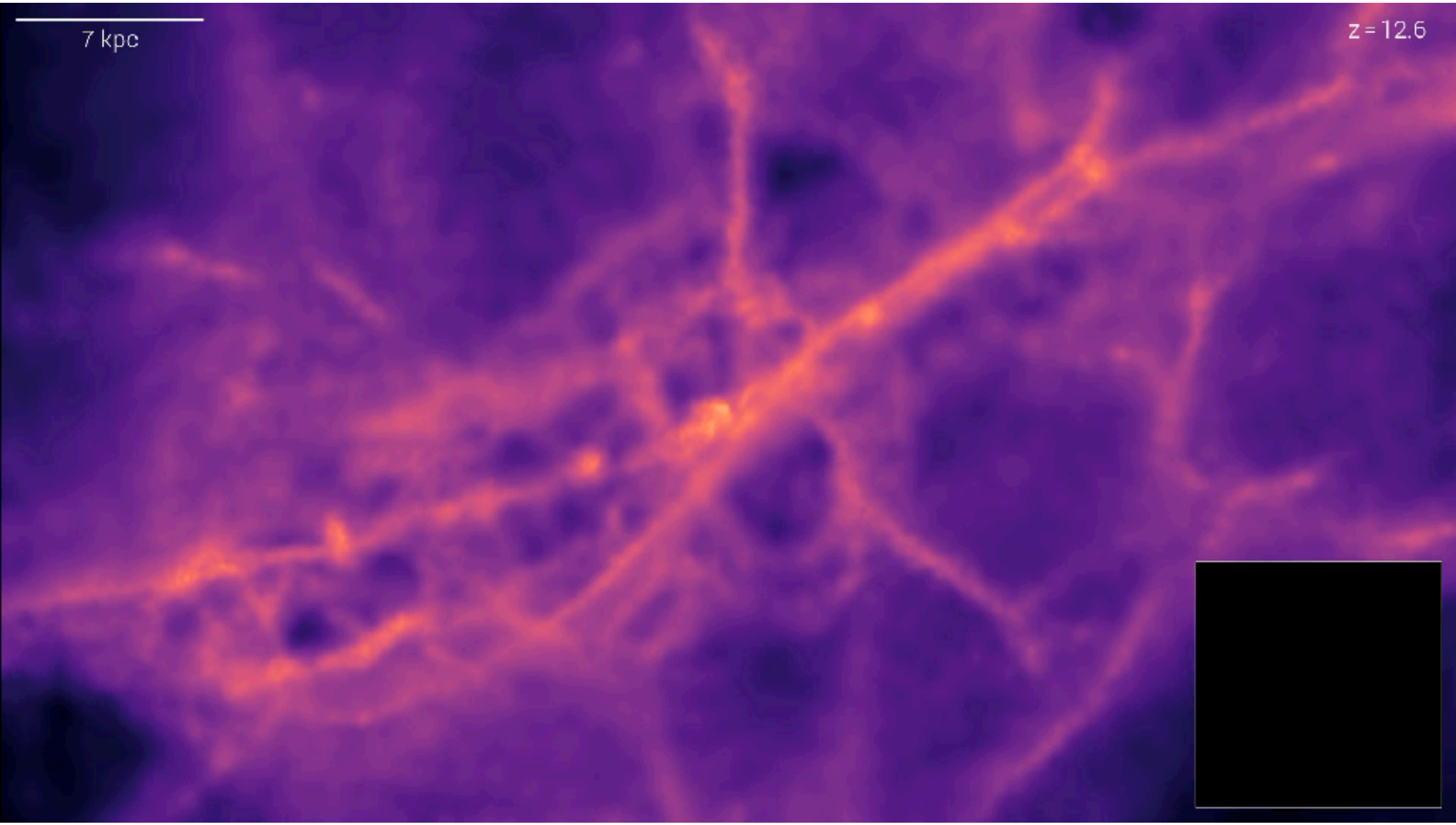
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Growth of SMBH and galaxies with cosmic time



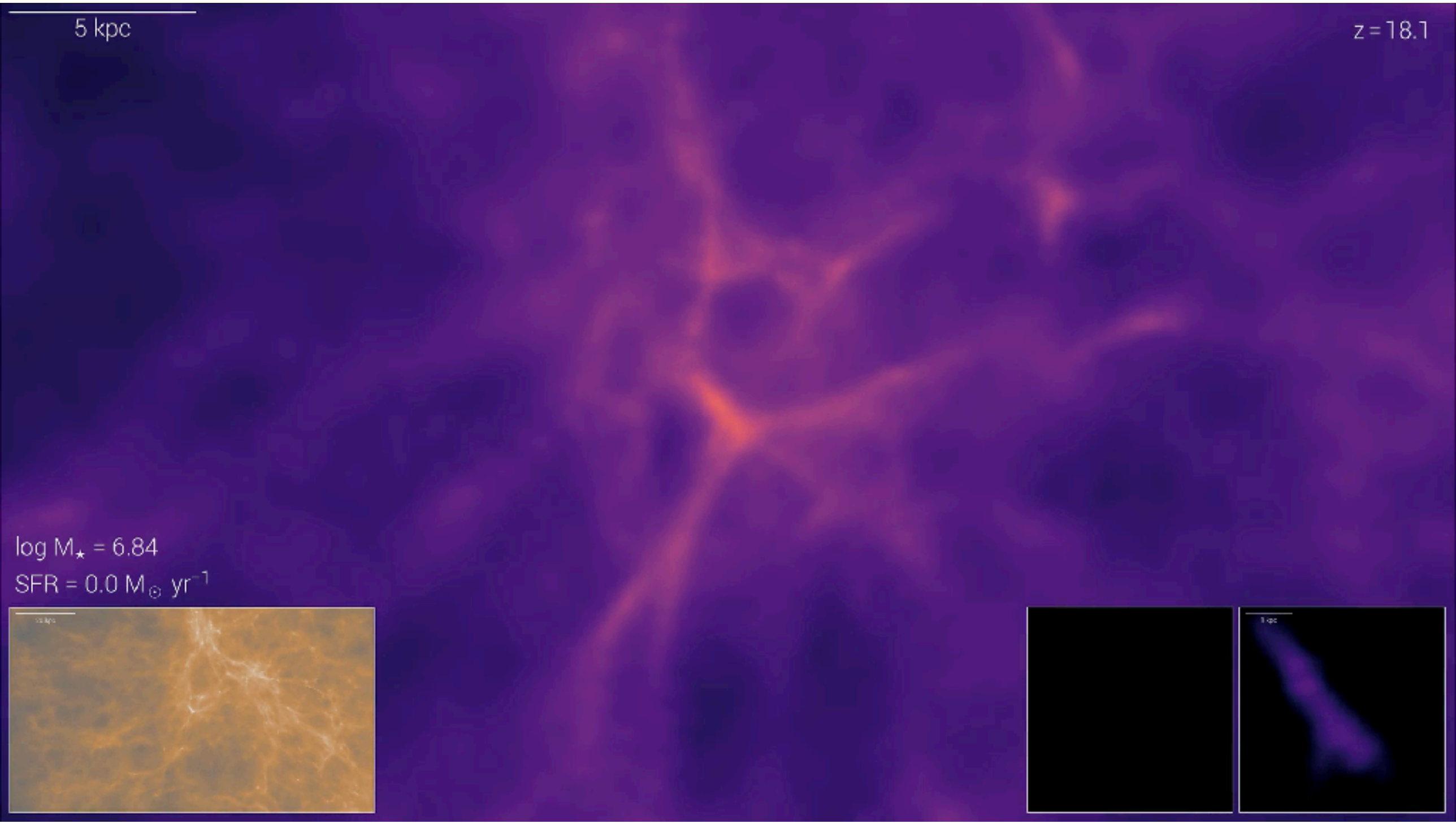
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Illustris TNG simulations



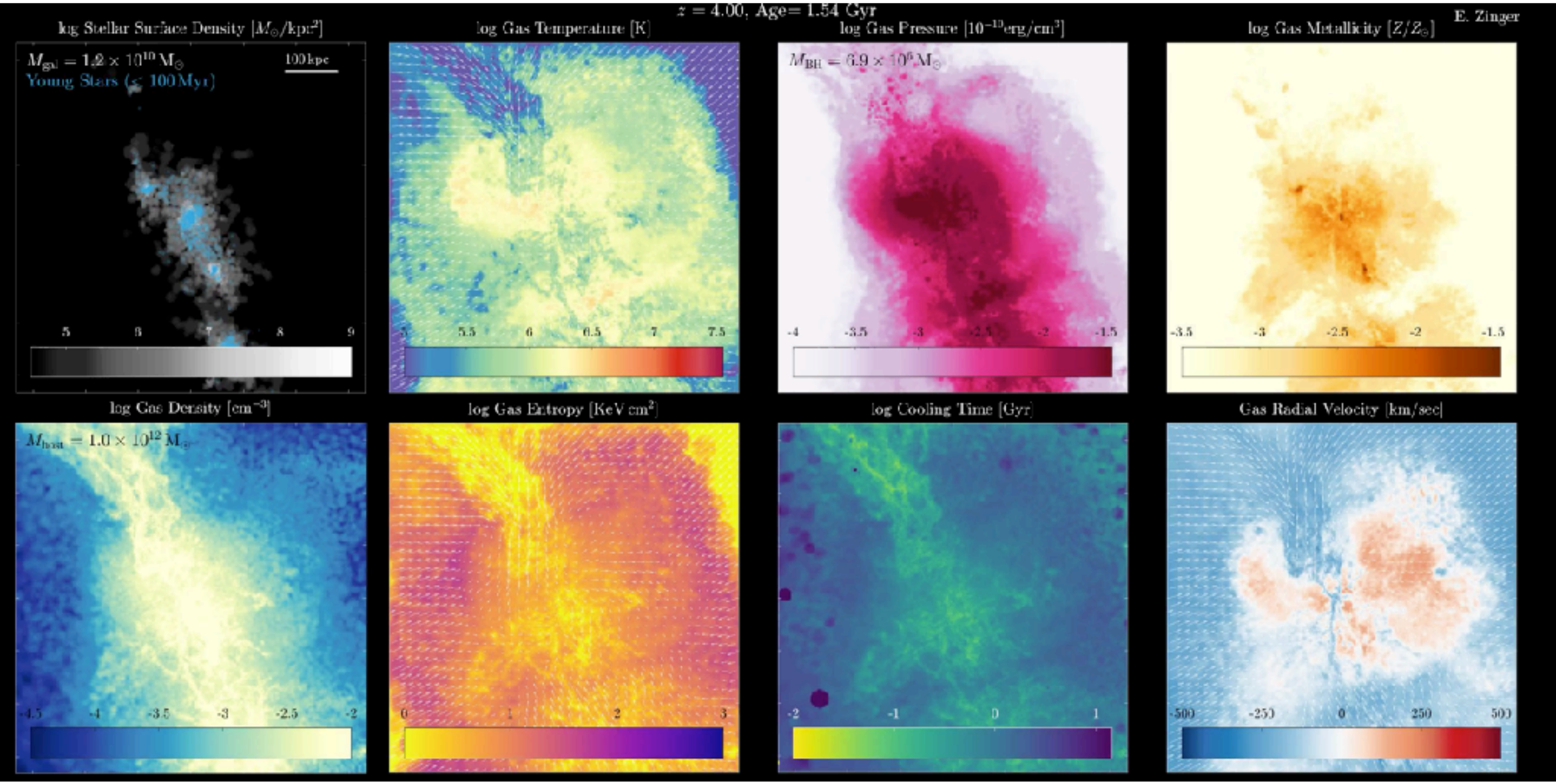
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Illustris TNG simulations



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Illustris TNG simulations



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Recent research on Huge Outflowing Galactic Winds

Being at an RI research university, people on campus are doing world-renowned cutting-edge research every day.

Let’s talk about how doing research really works and how discoveries work.

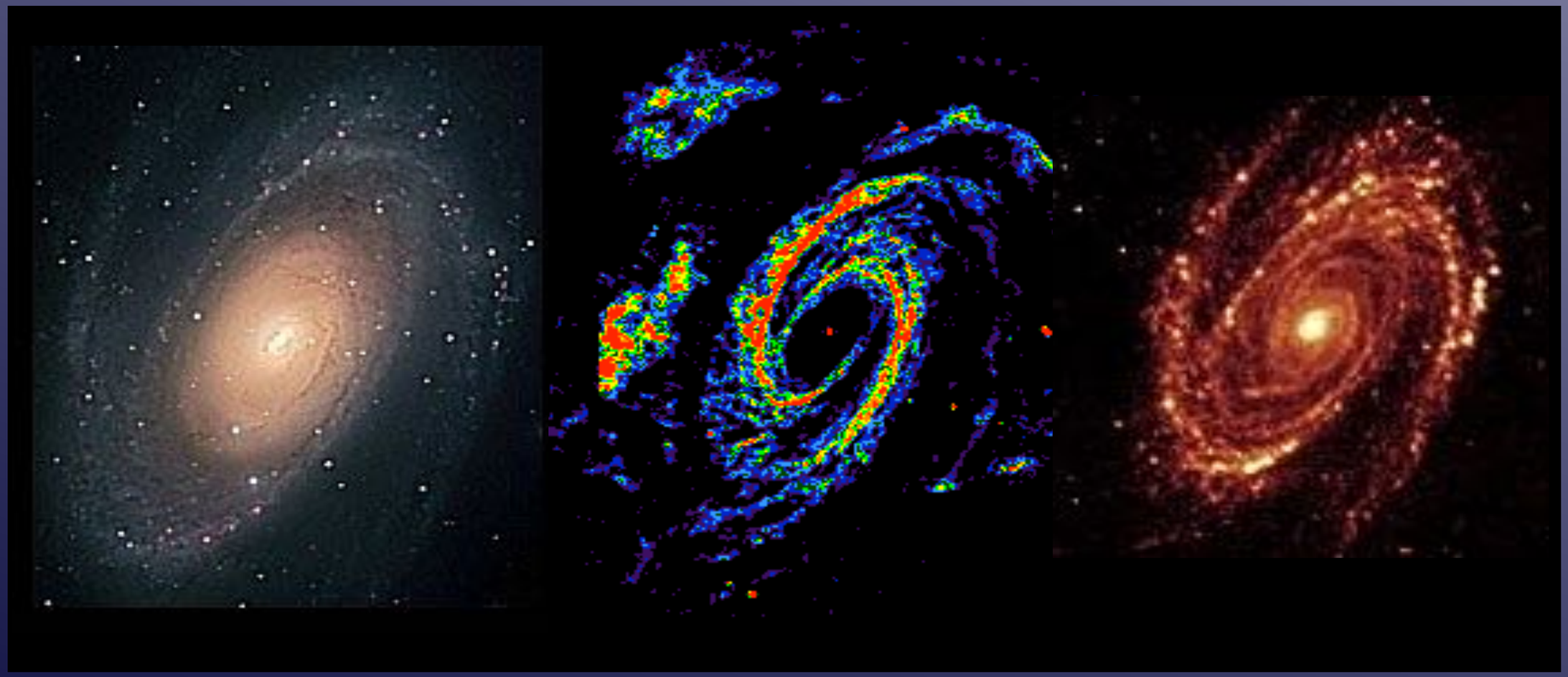
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What are galaxies made of?

Stars

Gas

Dust



Optical

Radio

Infrared

Where is all of the matter?

From what we know about the Big Bang, 5% of the mass/energy in the Universe today should be in the form of "normal" matter (baryons): protons, neutrons, electrons. This makes up the stars, gas, and dust that we see.

But when we count up the baryons we can see, *we can't find most of the baryons that should exist!*



Can only see half of the amount that should exist in stars, dust, planets, galaxies, etc.

Survey Question

What is the "missing baryon problem"?

A) The mismatch between the number of observed and predicted galaxies.

B) The failure to detect baryons in the early.

C) The mismatch between the observed and expected fraction of baryonic matter in the universe.

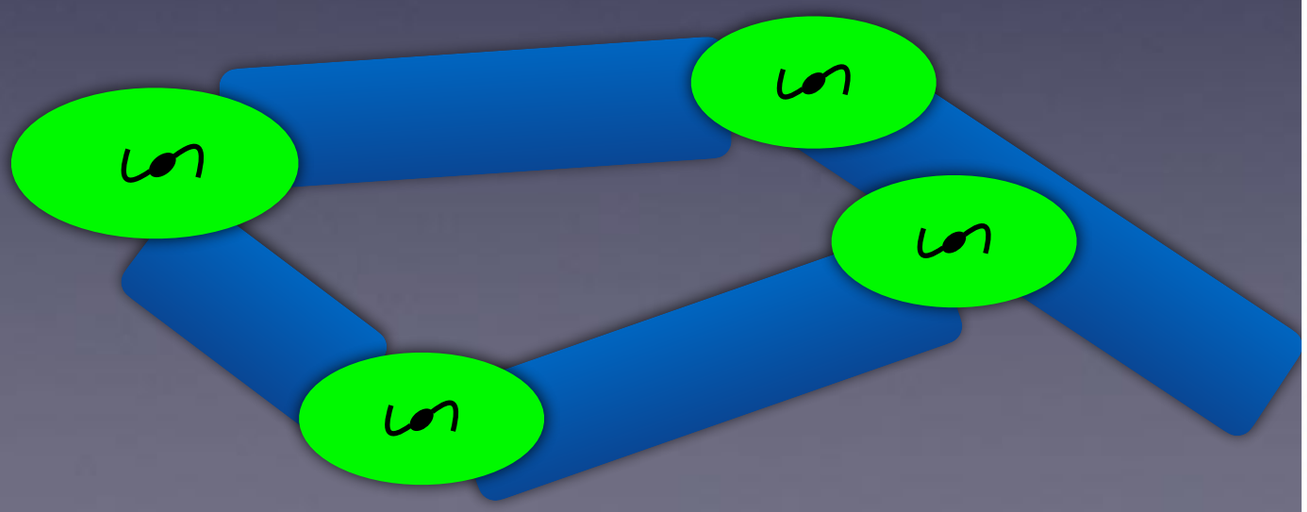
D) The reason we think dark matter exists.

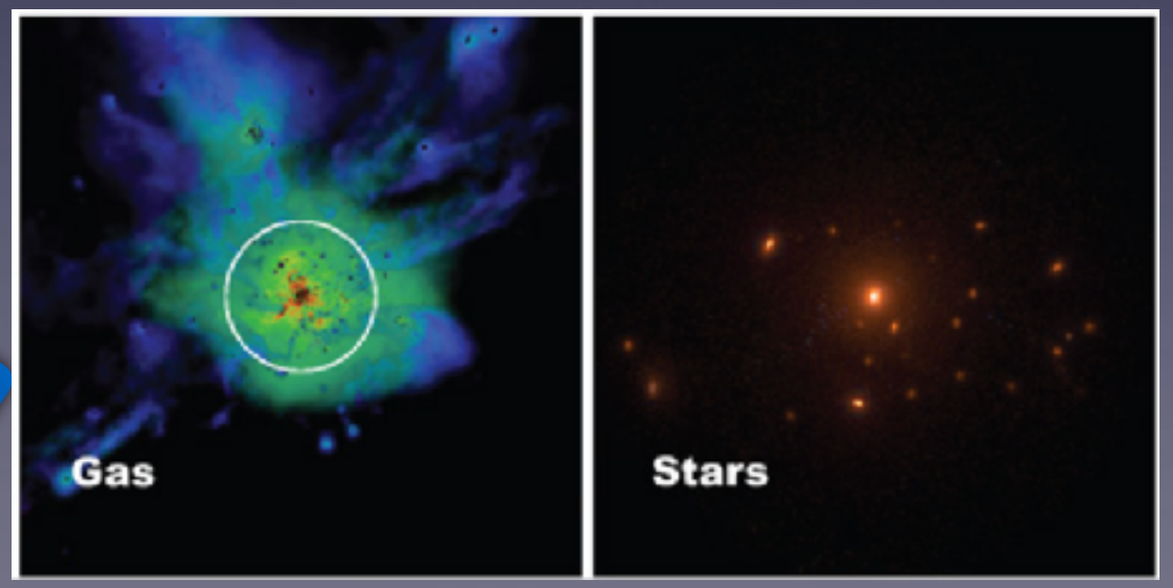
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Missing Baryons Problem

It turns out many baryons are around and in between galaxies in the circumgalactic medium (CGM, green) and intergalactic medium (IGM, blue).

Very hard to detect due to low gas density and multiple gas phases (cold, warm, hot).





Circumgalactic Medium

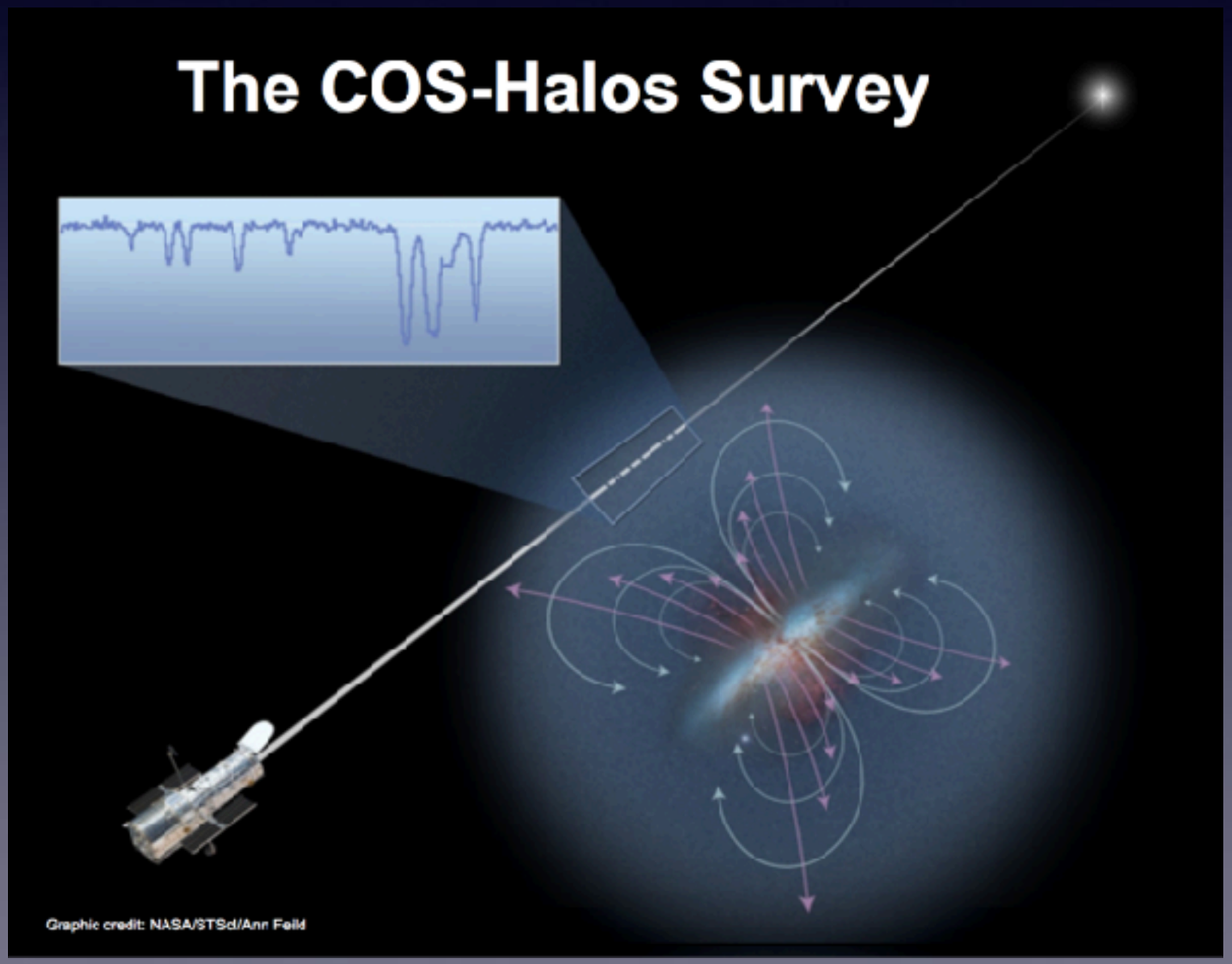
The circumgalactic medium (CGM) is the gas surrounding galaxies that is *outside* the stellar disk of the galaxy but is still gravitationally bound to the galaxy.

It is the reservoir of gas needed to fuel future star formation!

We know that there is a lot of mass in the gas in the CGM, and that is has multiple temperatures and phases.

Circumgalactic Medium

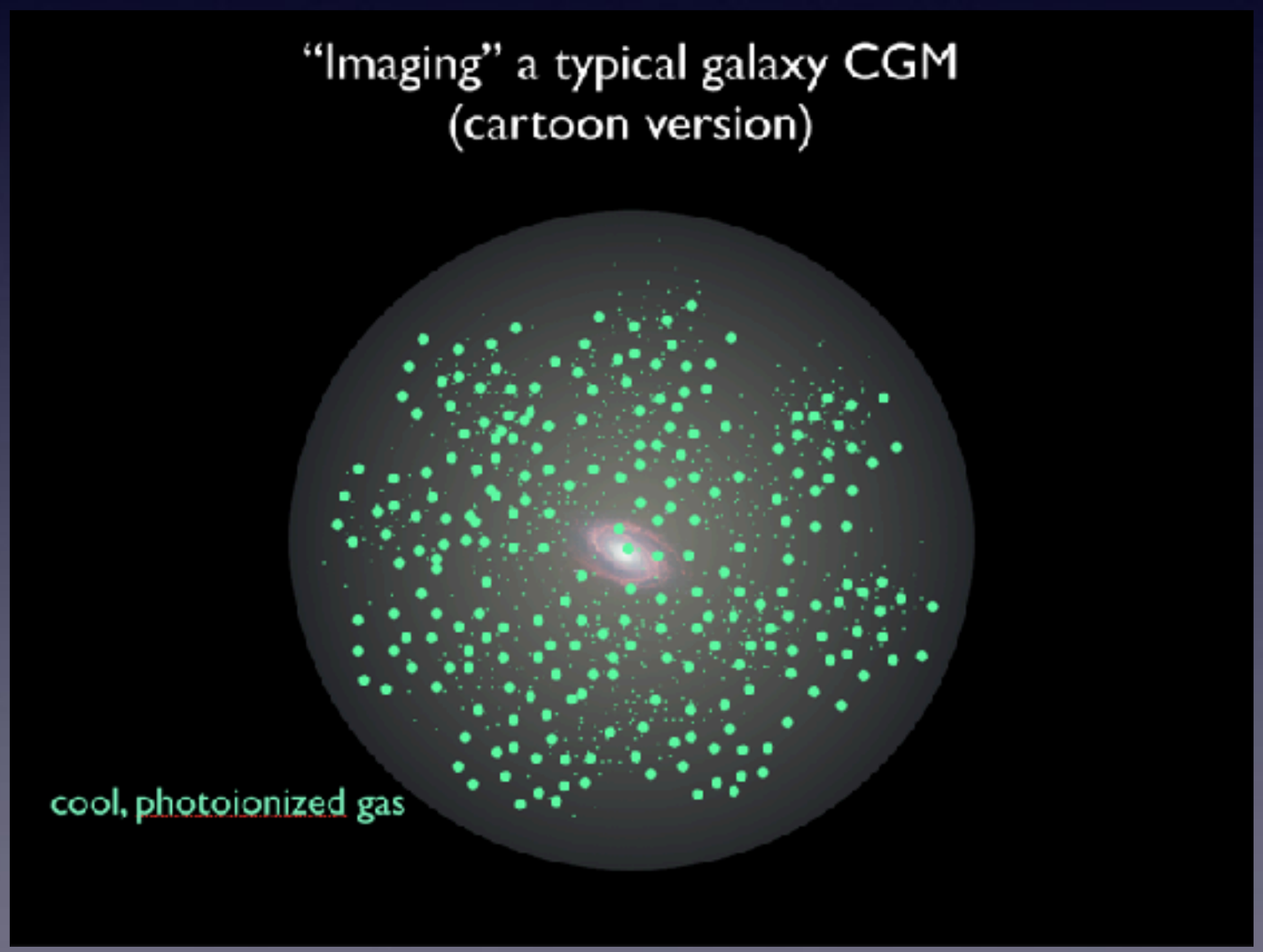
Observationally, we probe the CGM through absorption sightlines to background quasars.



A lot of mass in the CGM - more than in the stars, gas, and dust in the disk!

Circumgalactic Medium

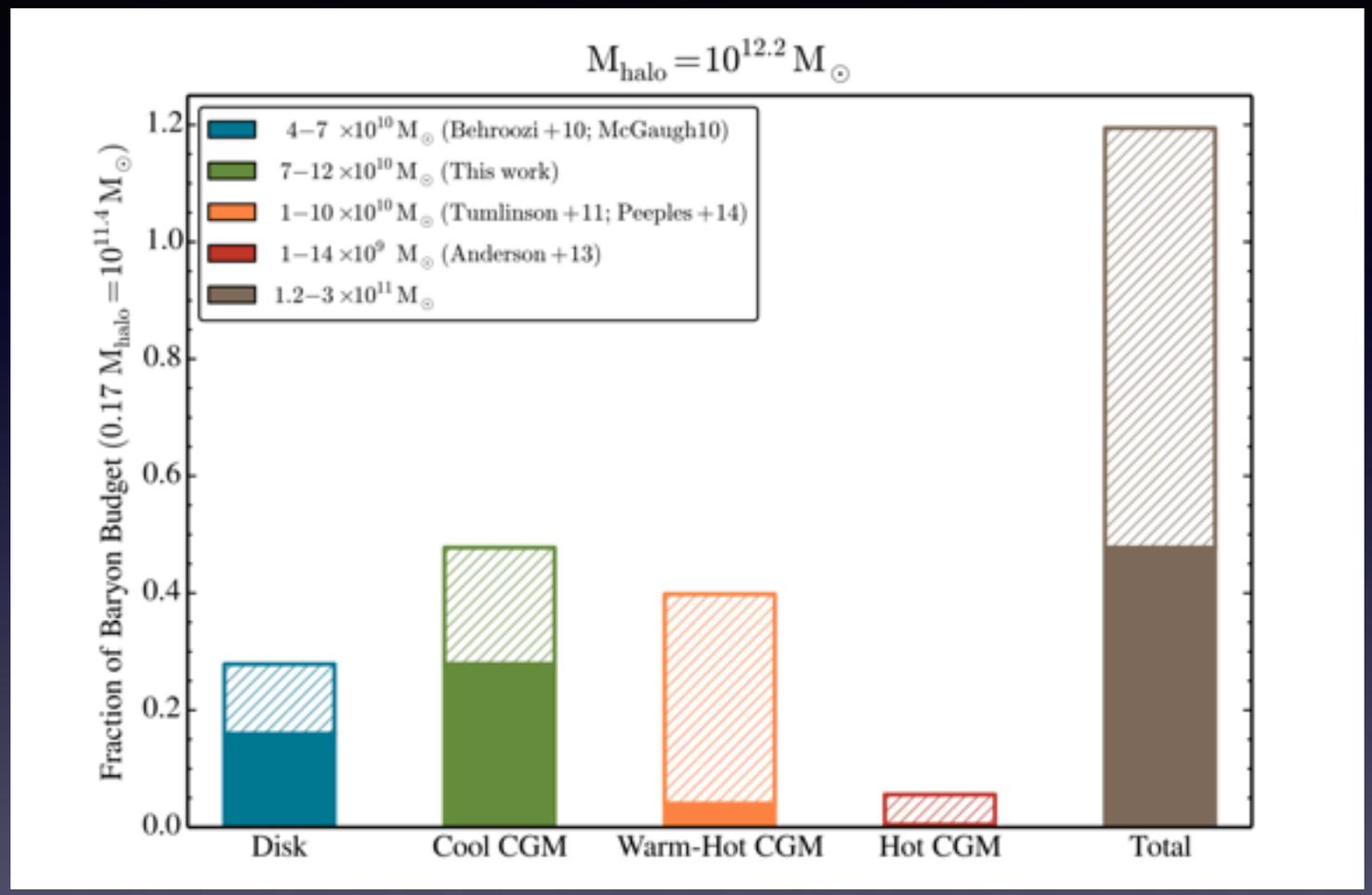
Statistically, we can build up a picture of what the CGM must look like, on average, for galaxies:



A lot of mass in the CGM - more than in the stars, gas, and dust in the disk!

Circumgalactic Medium

Baryon budget of galaxies



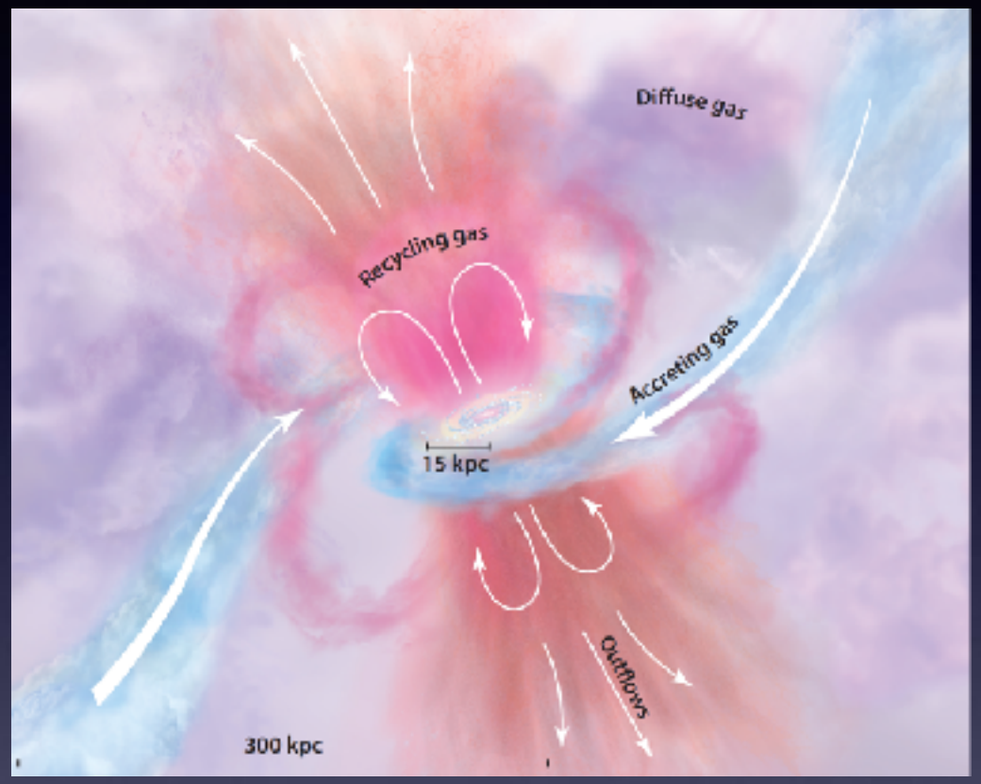
Werk et al. (2014)

From sightlines around 44 galaxies, find more baryons in the CGM than in the galaxy disk!
Most of the gas is in a cool phase ($T \sim 10^4$ K).

Origin of the Circumgalactic Medium

Theory points to the CGM being created both by:

- infalling gas from the cosmic web
- outflowing gas from galaxies



Tumlinson, Peebles & Werk (2017)

Observationally, we have statistical knowledge of the CGM by detecting it in absorption across <u>different</u> halos and <u>different</u> sightlines. Don't get kinematic information about the gas relative to the galaxy that way (ie, infalling or outflowing) or morphology.

Nearby Galactic Outflows



M82, nearby starburst galaxy, optical image

Nearby Galactic Outflows



M82, infrared image showing cool gas and dust

Nearby Galactic Outflows



M82, optical (green) + IR (red) + X-ray (blue)
outflowing gas and dust on similar scale of disk

Nearby Galactic Outflows

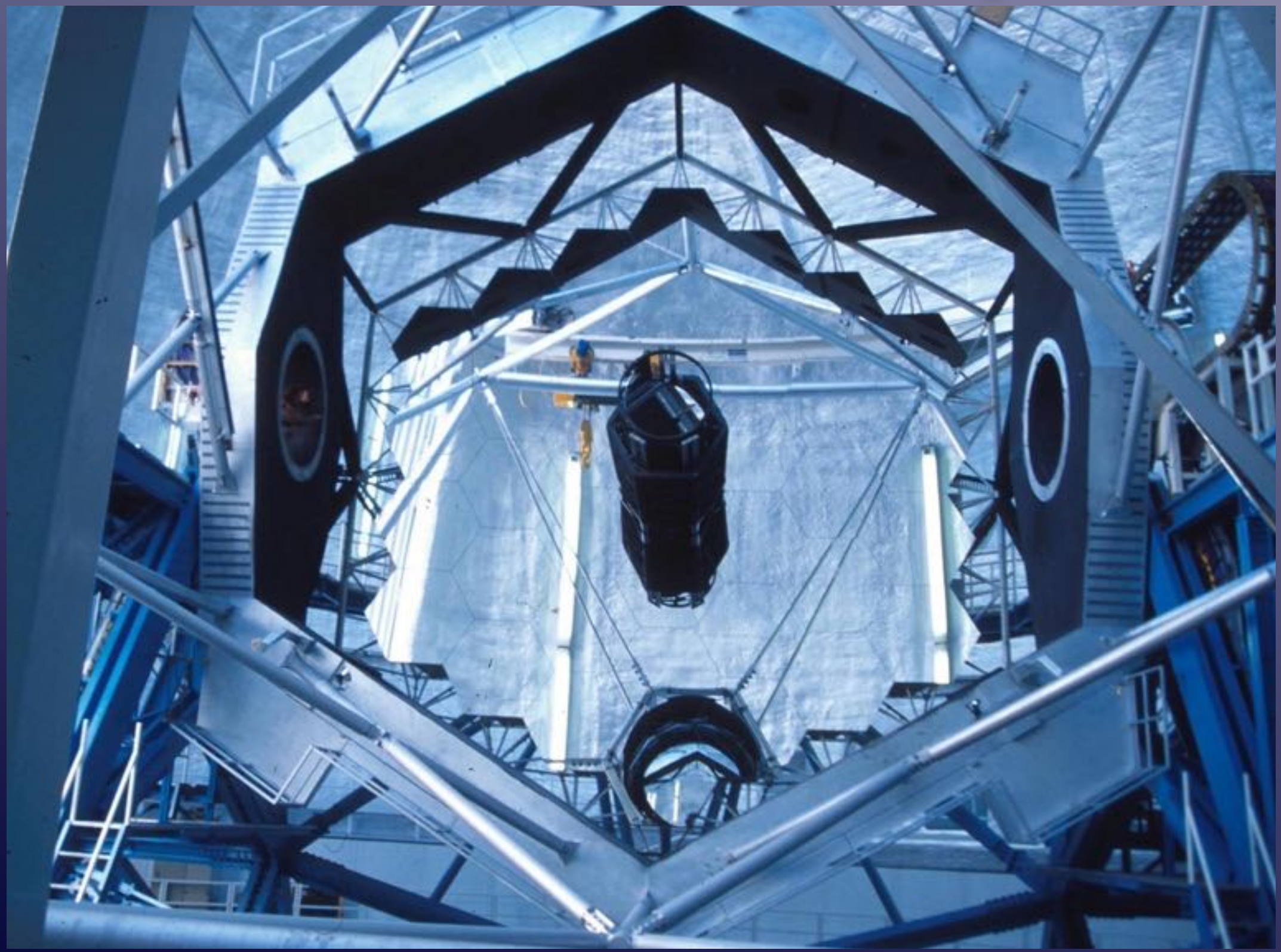


M82, only 11 million light years away

Distant Galactic Outflows

- We can't see this kind of dust and gas in emission easily in distant galaxies, billions of light years away, as it's too faint.

Keck 10-m telescope



Detecting Inflows and Outflows

"Down the barrel" observations towards a distant galaxy

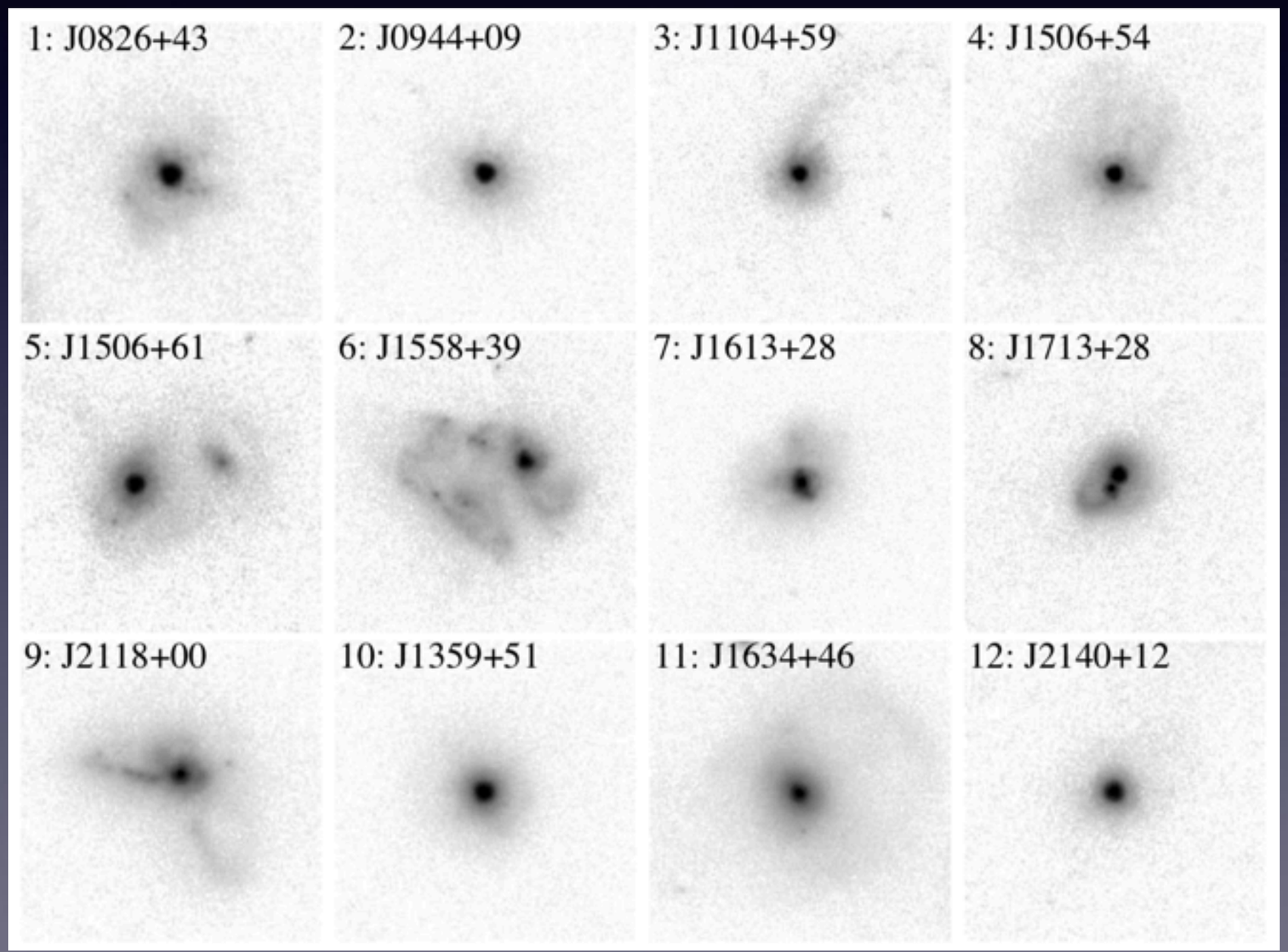
Inflows: gas is redshifted relative to gas/stars in galaxy



Outflows: gas is blueshifted relative to gas/stars in galaxy



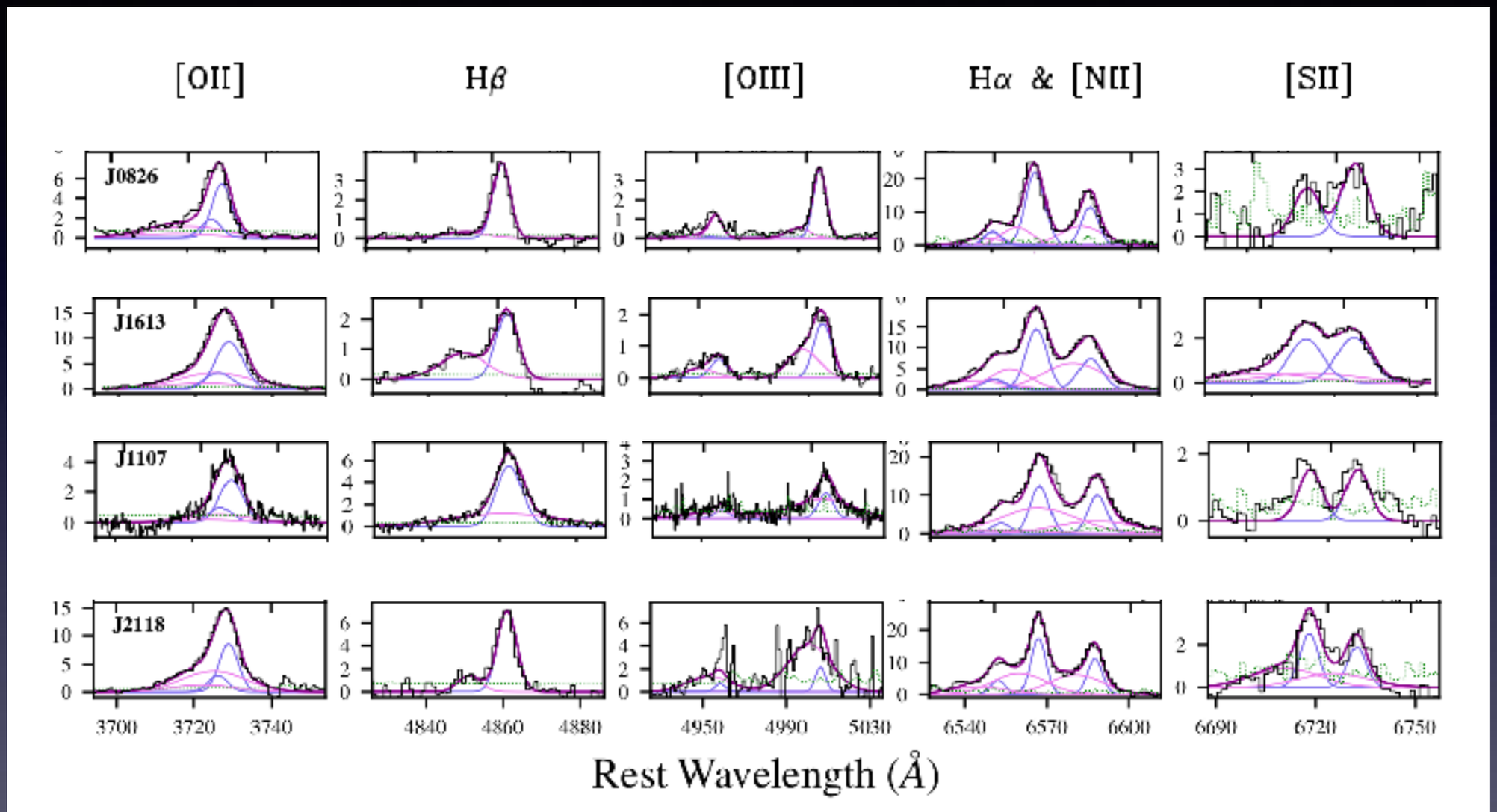
Distant starburst galaxies with extreme outflows



Science is not linear - lots of twists and turns

- Every time we would point a telescope at these galaxies, we learn something new and unexpected about them!
- They don’t have AGN!
- They are not post-starburst but are starburst!
- They have huge molecular outflows!

Outflowing winds in emission



Perrotta, Coil, et al. (2021)

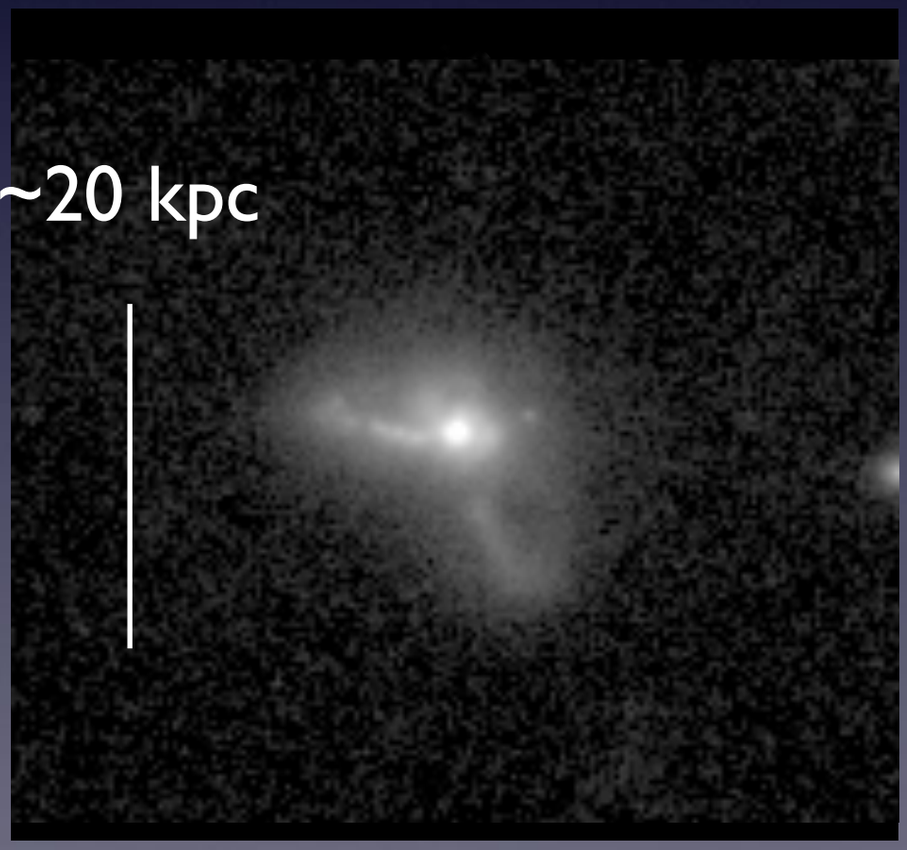
Keck/NIRSPEC + MMT data

see additional broad blueshifted emission (pink)

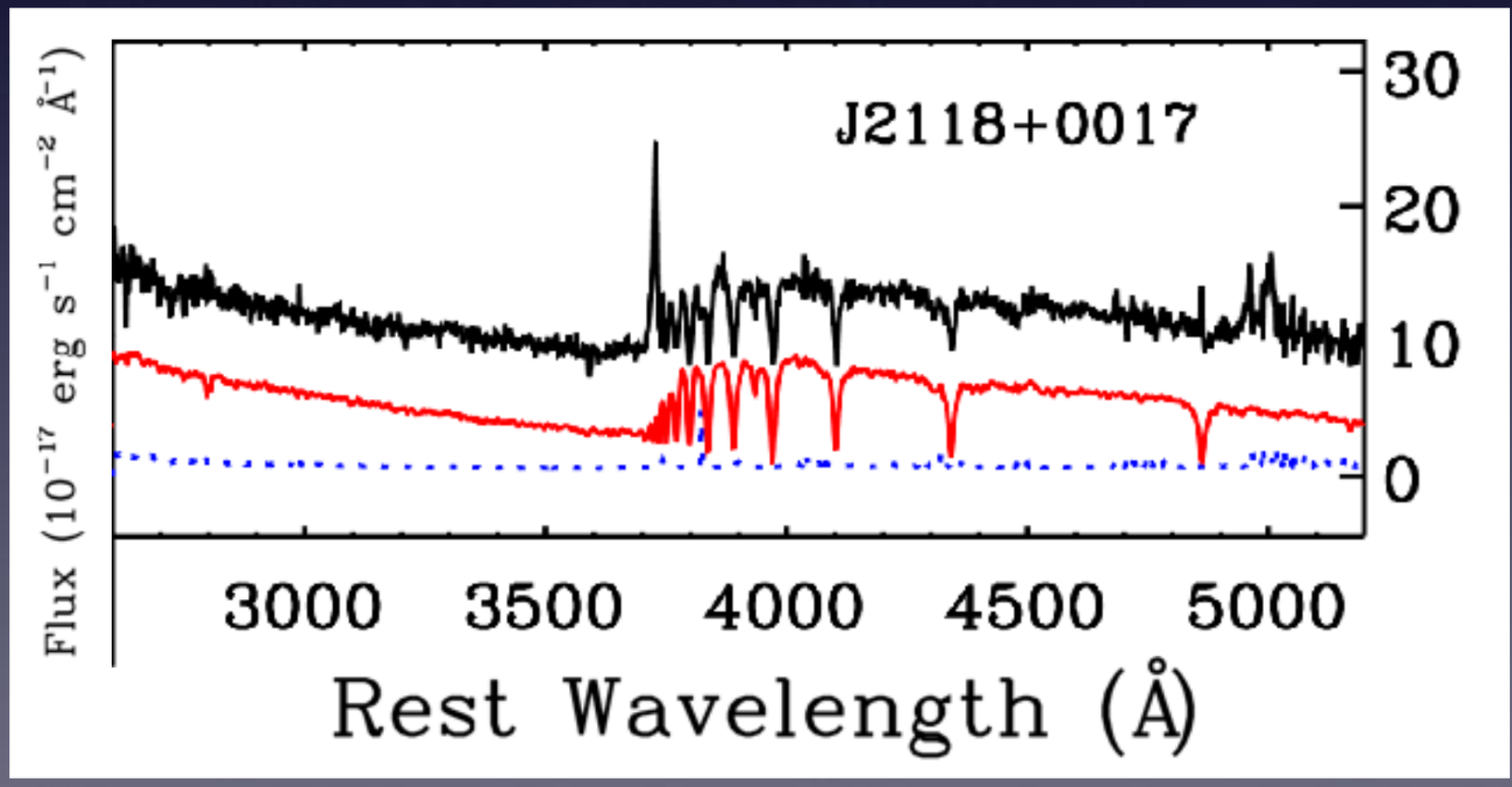
J2118+0017

z=0.459, log M/M* = 11.1 (+/-0.2), SFR~150 M/yr

HST: highly peaked stellar core (radius 400 pc) with tidal tails, half-light radius 2.5 kpc



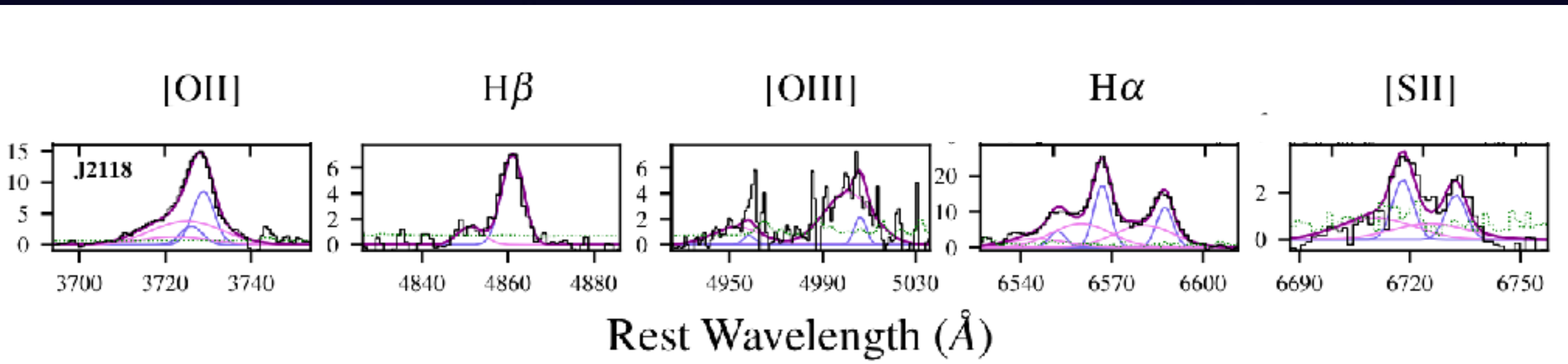
HST WFC3/F814W



MMT + SDSS spectra

J2118+0017

SDSS + NIRSPEC spectra of restframe optical lines:



Additional broad, blueshifted emission - very strong in [OIII].
No Mg II absorption - we originally didn't know if it had a wind.

Survey Question

Which of the following suggests gas is flowing into or out of a galaxy?

A) The presence of emission and absorption lines in a galaxy's spectrum.

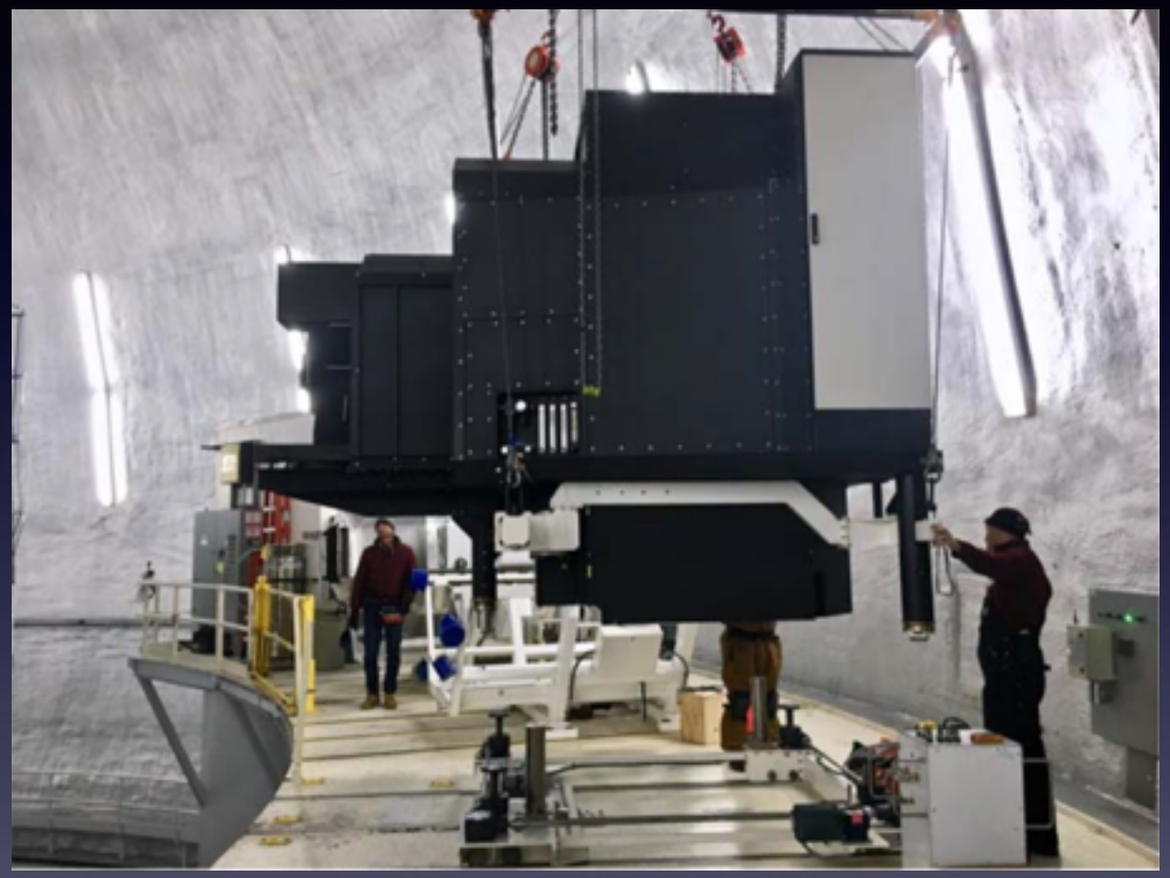
B) Emission lines that are redshifted or blueshifted relative to the galaxy's stars.

C) Absorption lines that are redshifted or blueshifted relative to the galaxy's stars.

D) Both B and C.

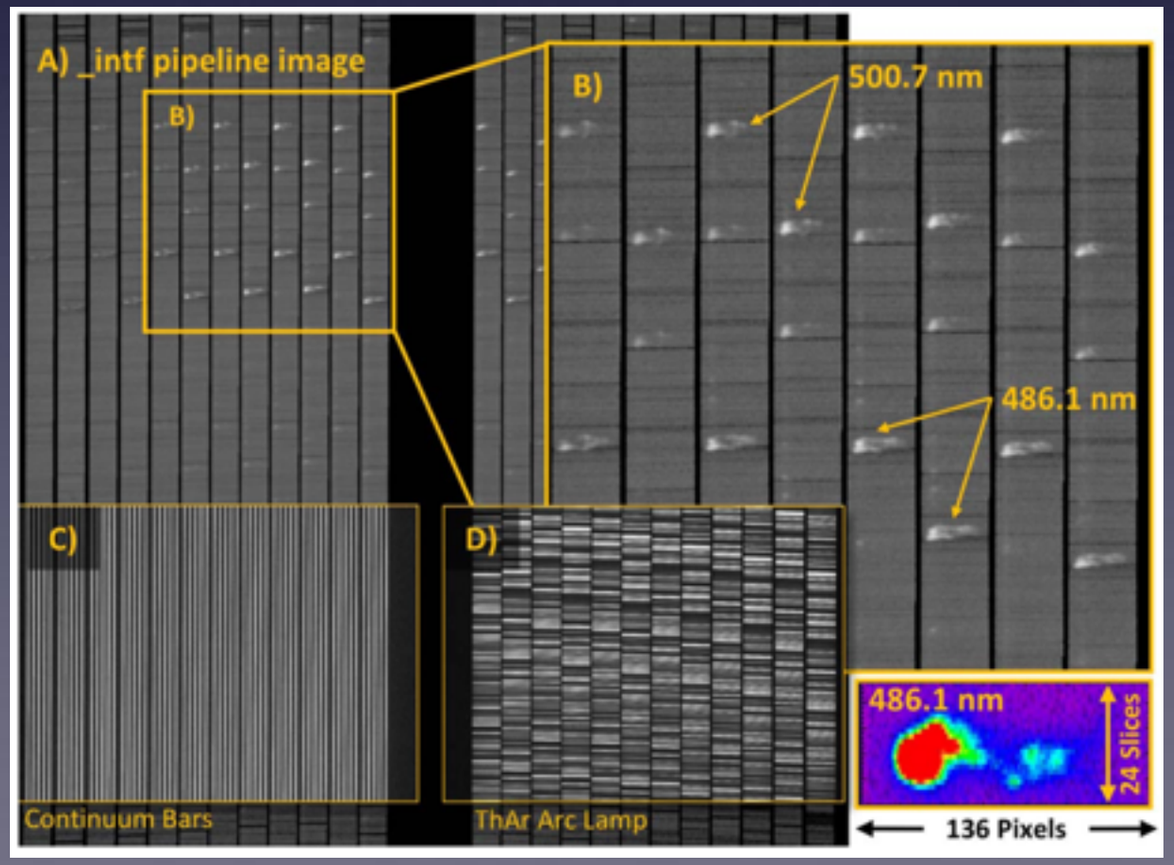
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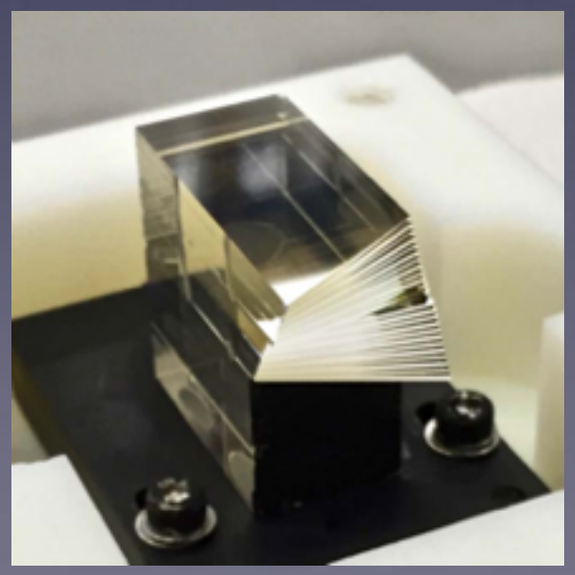
KCWI: Keck Cosmic Web Imager



Integral field spectrograph

Sensitive to very low surface brightness emission





J2118+0017

$M_* = 10^{11.1}\ M_\odot$

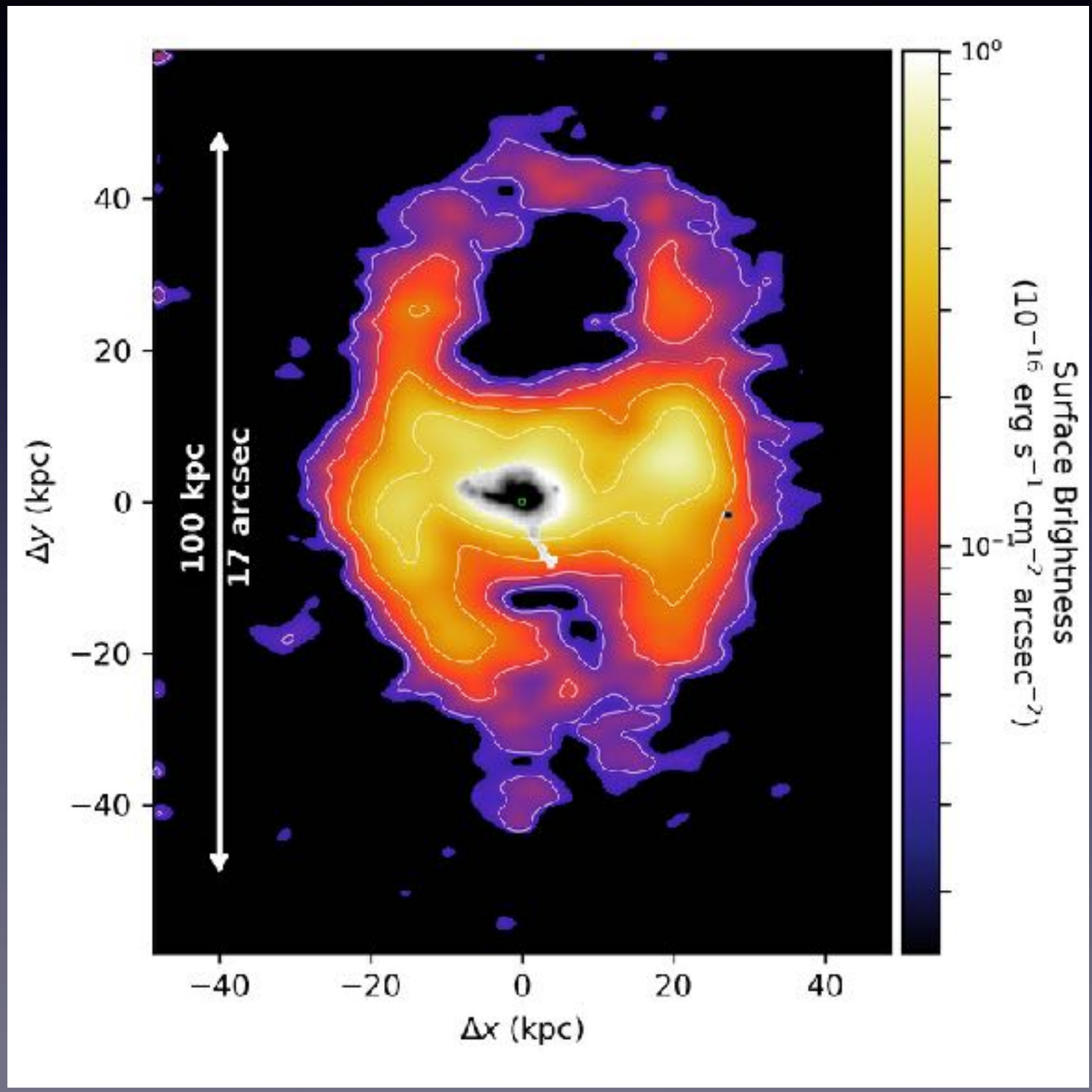
$z = 0.459$ merger

Makani
("wind" in Hawaiian)

[OII] emission

KCWI

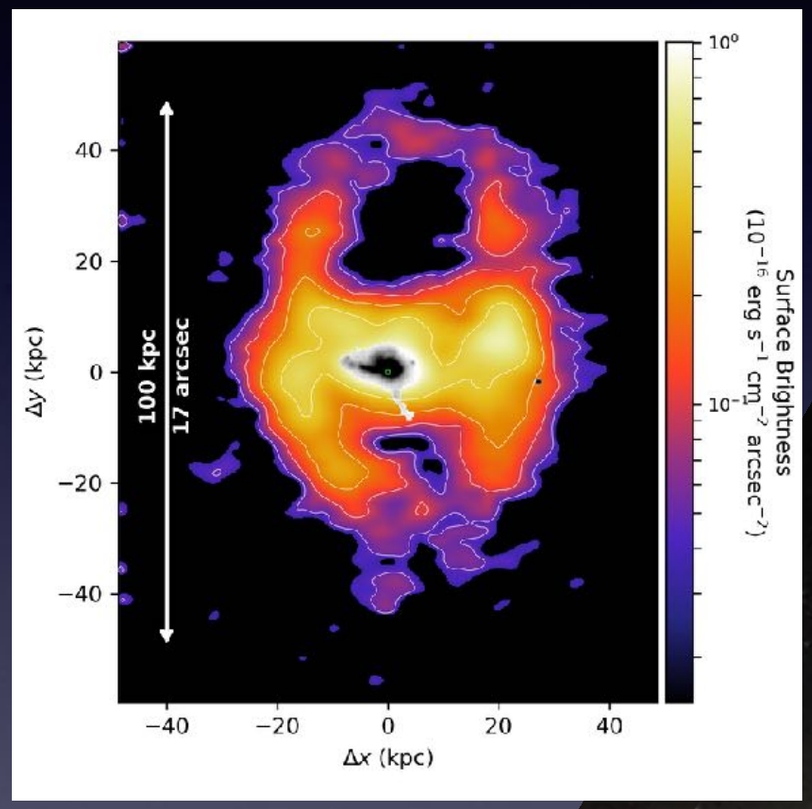
40 min. exp!



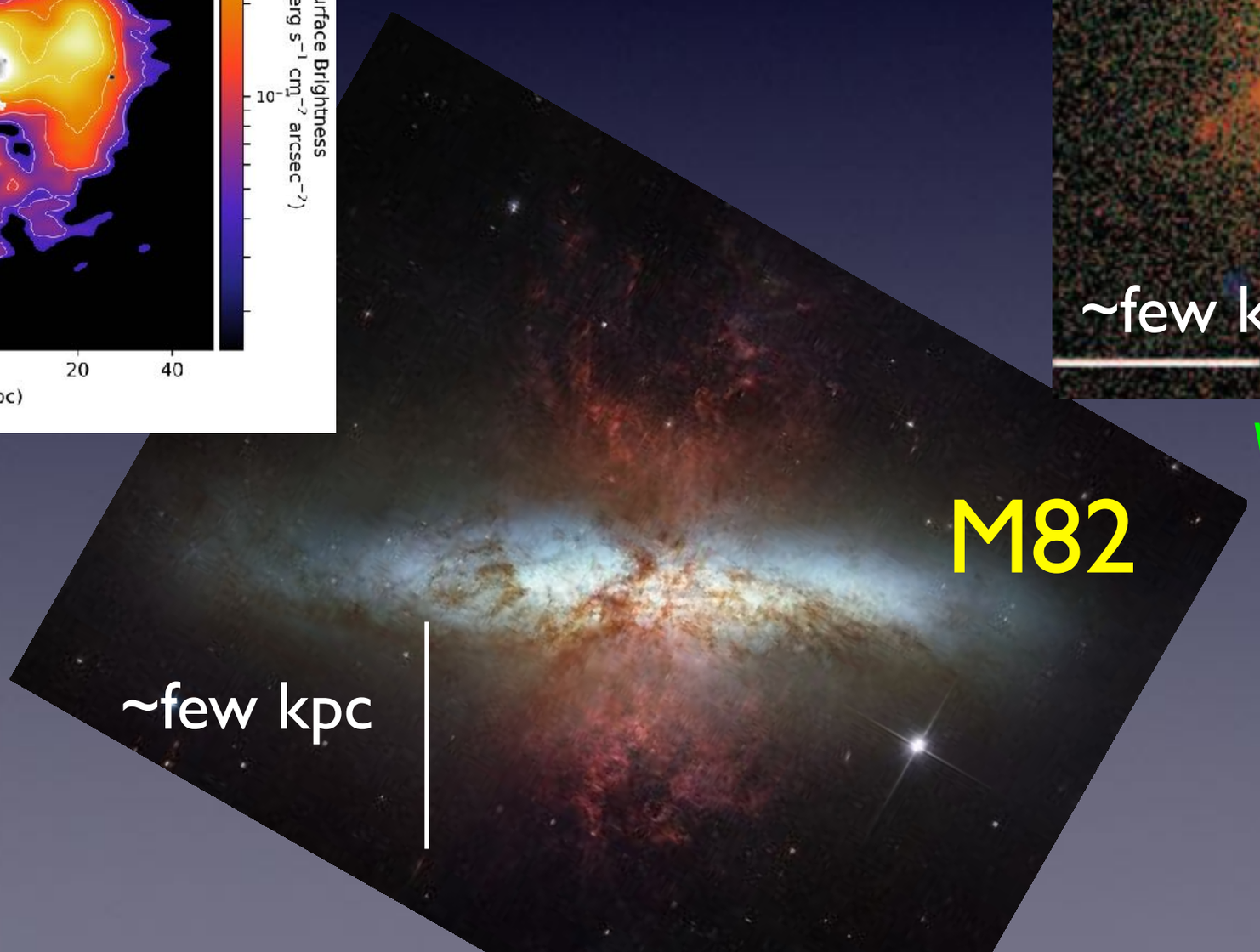
Rupke, Coil et al. (2019)

Classic hourglass limb-brightened bipolar bubble morphology, order of magnitude larger:

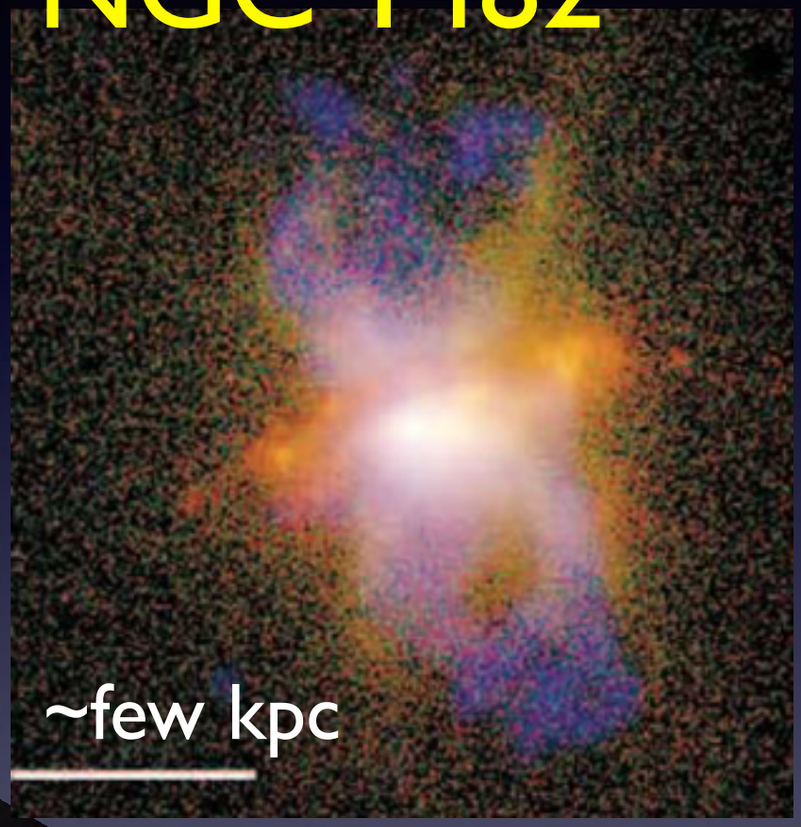
Makani



~100 kpc



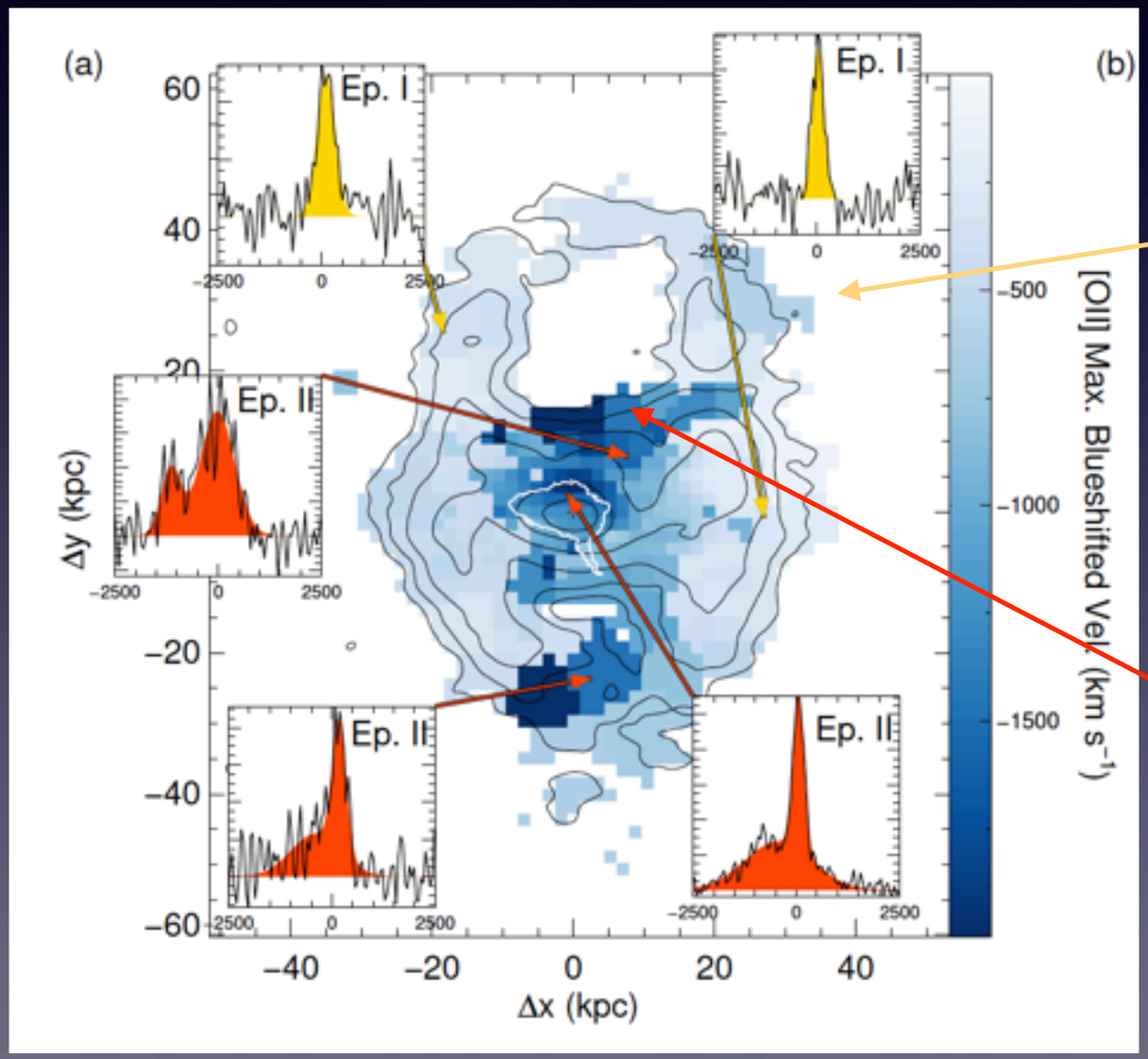
NGC 1482



Veilleux & Rupke (2012)

[OII] Kinematics

maximum blueshifted velocity

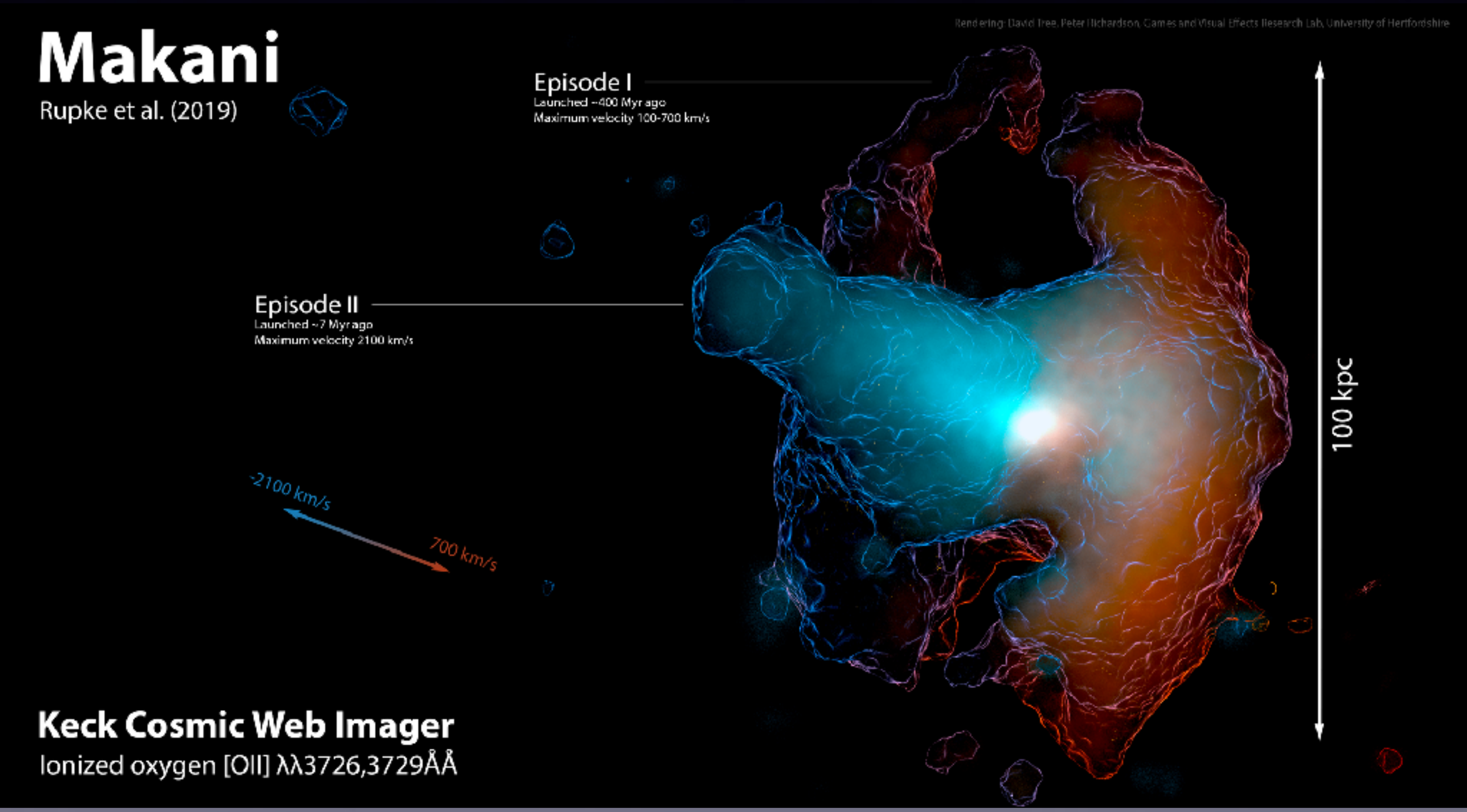


~ 300 km/s

~ 1800 km/s

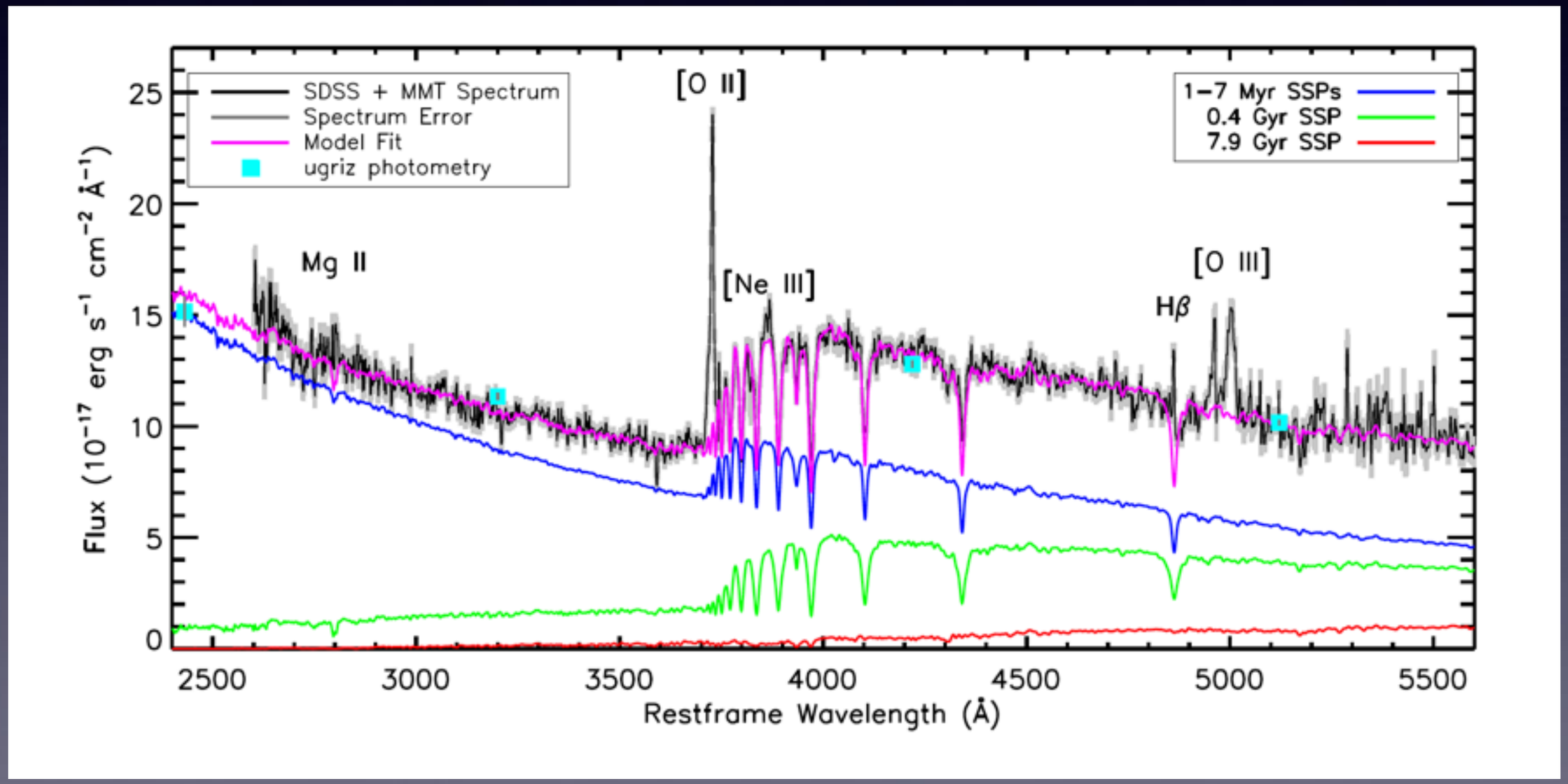
Rupke, Coil et al. (2019)

3D rendering, with velocity as z axis:
higher velocity is “further” away
(not the real 3D morphology!)



Stellar Population Fitting

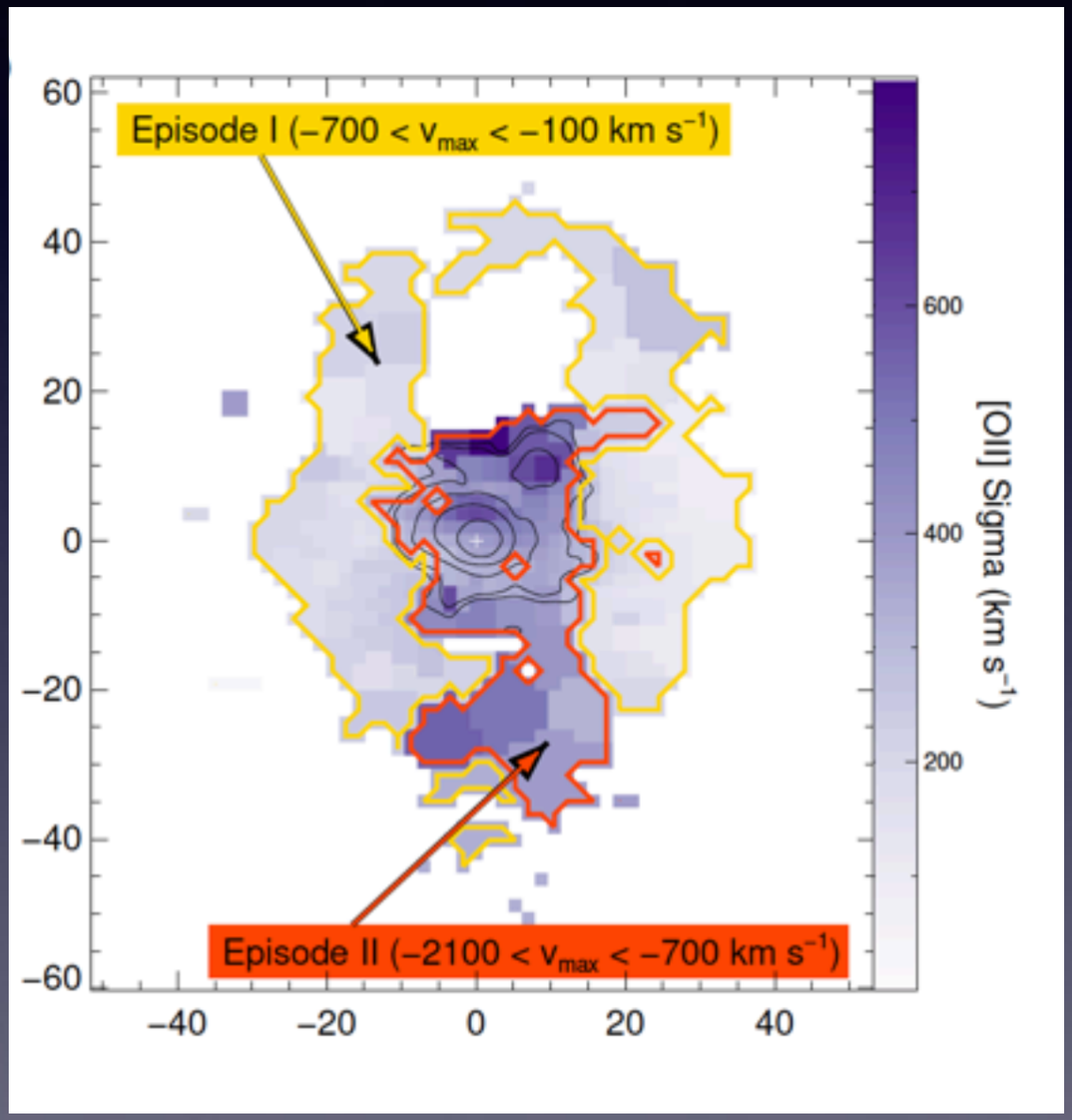
stellar fit shows 2 recent starbursts: <7 Myr and 0.4 Gyr ago



Rupke, Coil et al. (2019)

2 Outflow Episodes

[OII] kinematics:



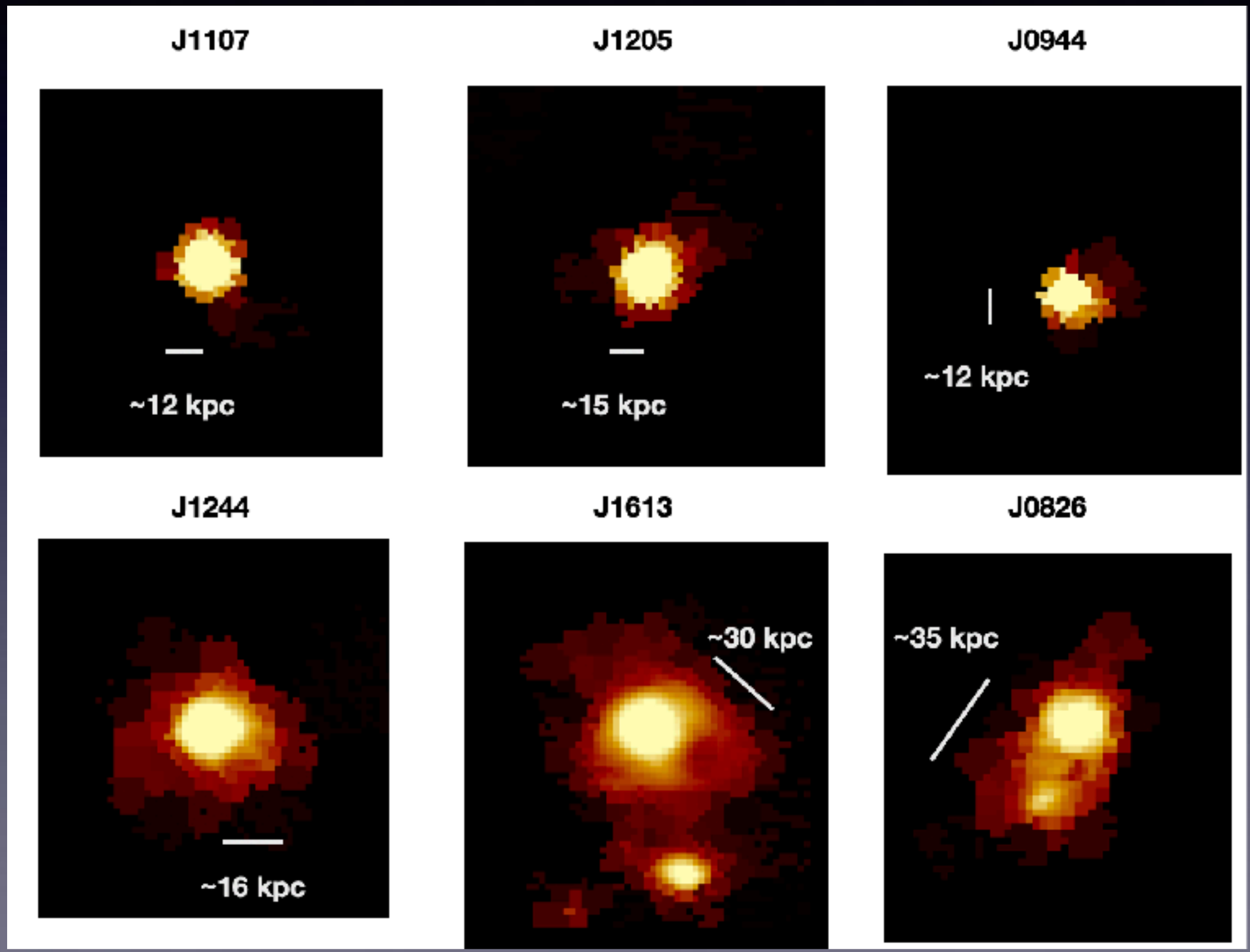
2 episodes:

- first: 400 Myr ago
  - gas moving at 120 km/s
  would have reached 50 kpc

- second: 7 Myr ago
  - gas moving at 1400 km/s
  would have reached 10 kpc

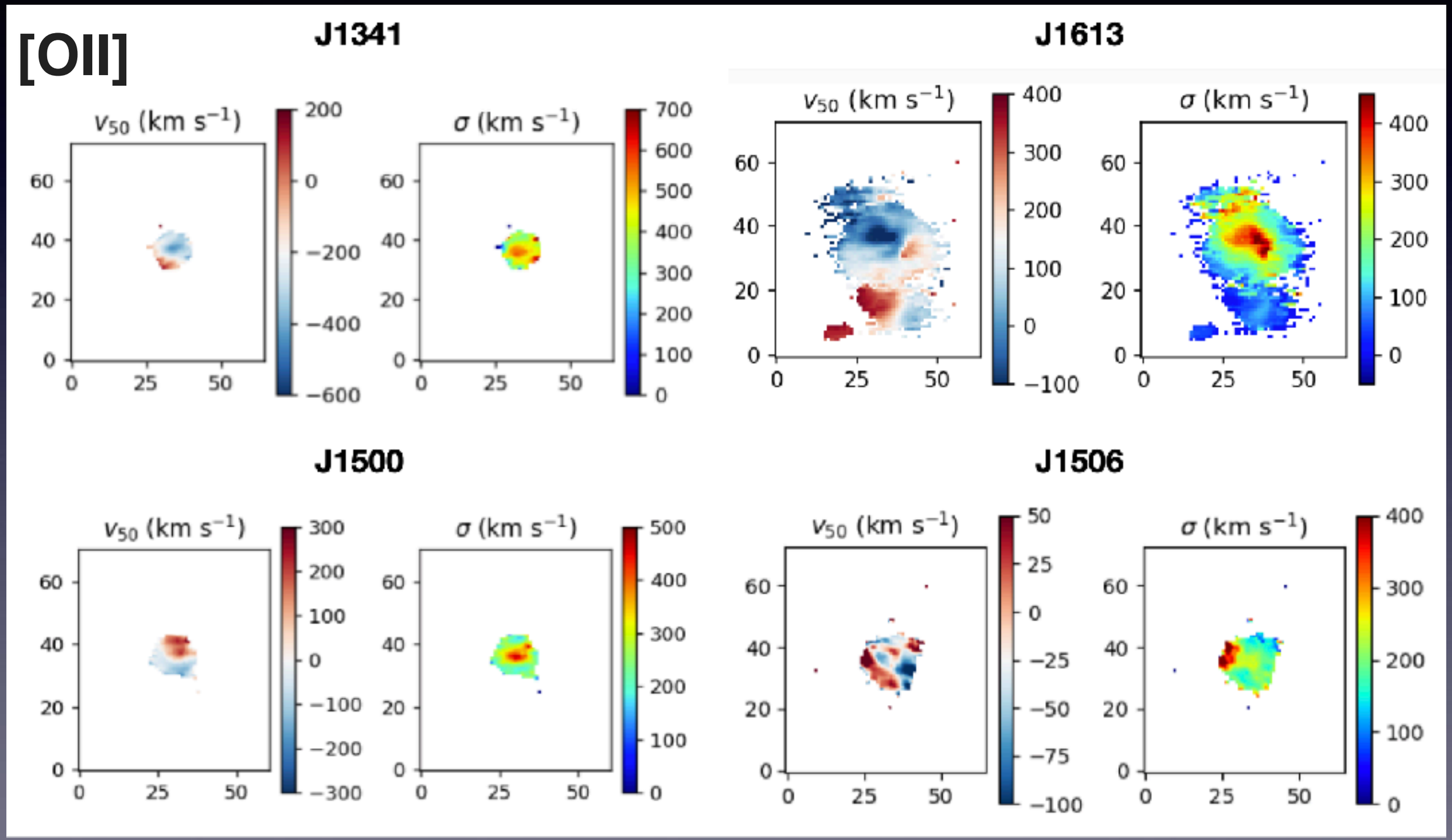
Speeds and sizes of the two outflows are consistent with being driven by the two known starburst episodes!

KCWI data on more galaxies!

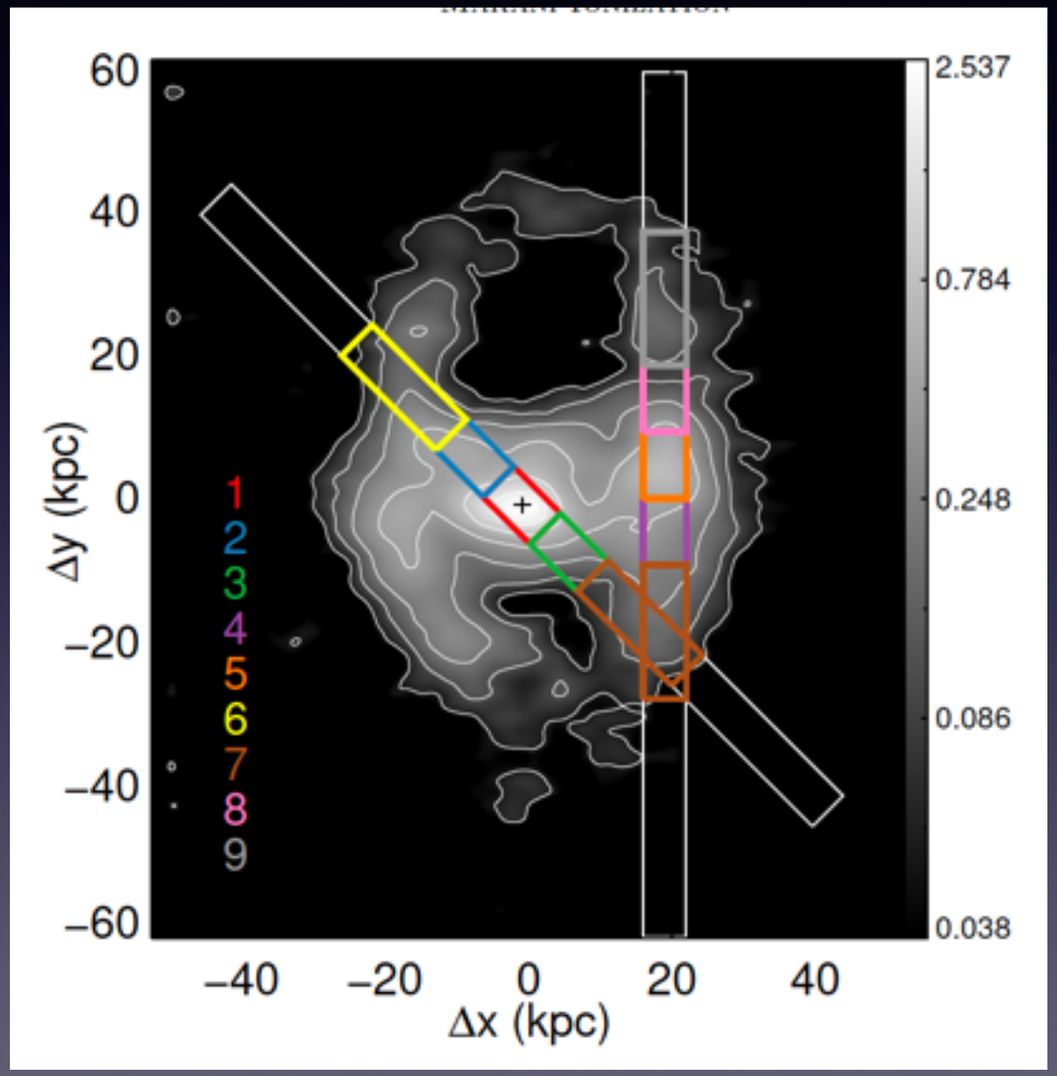


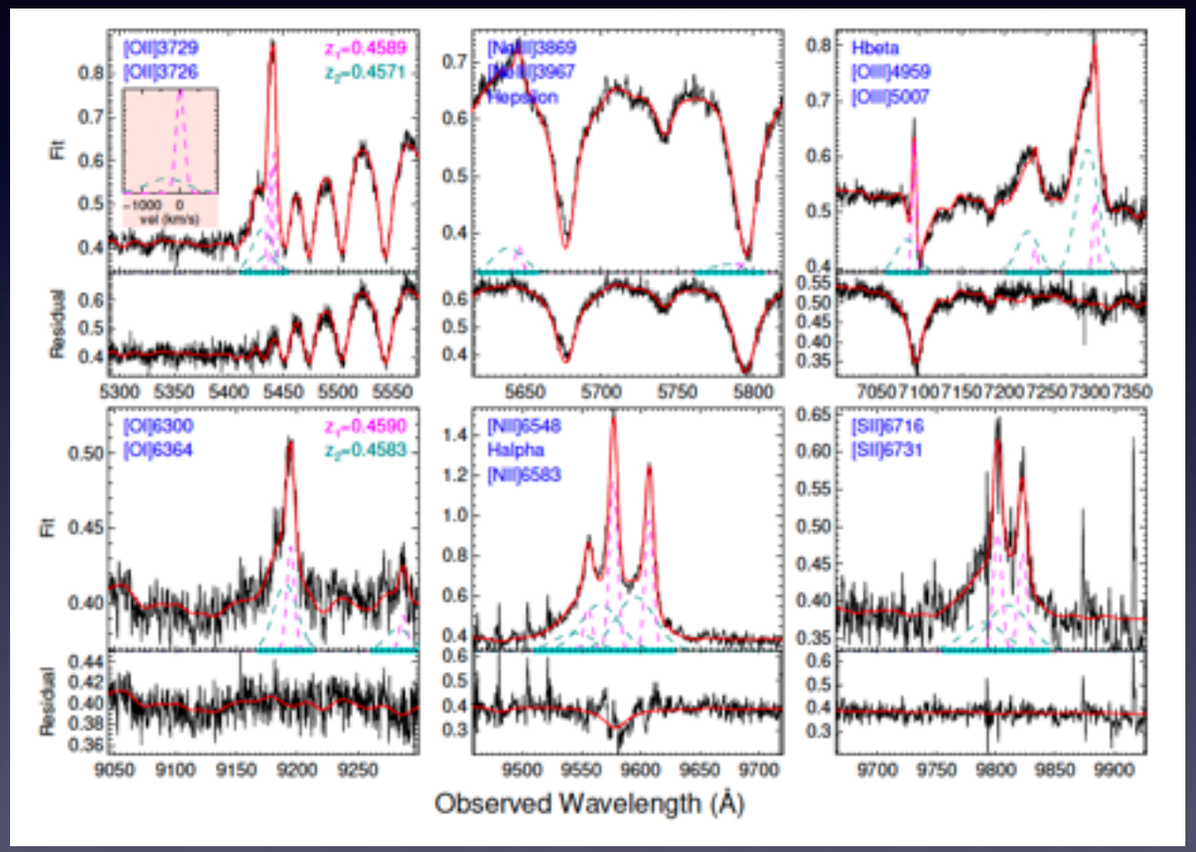
[OII]

Rich velocity information



Gas Mass in the Outflows





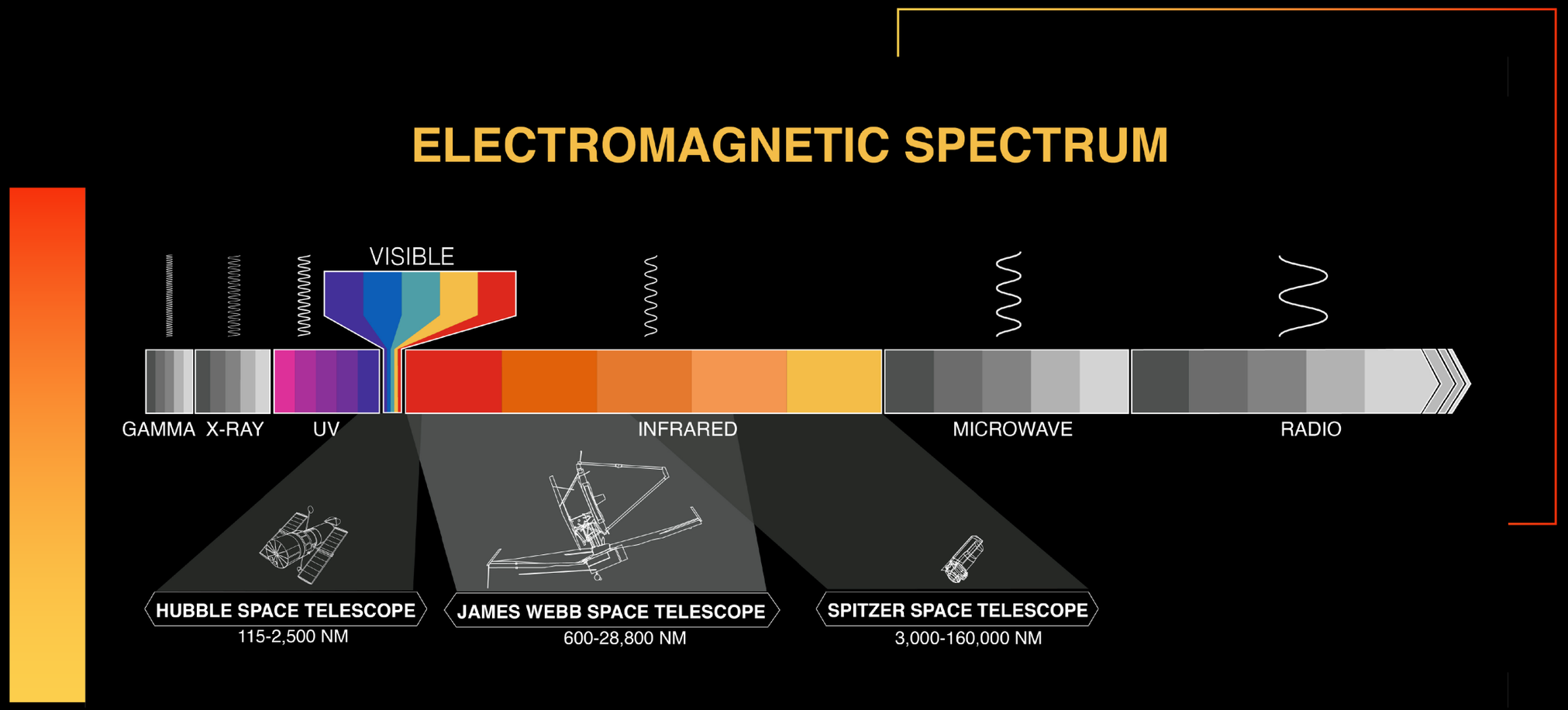
Rupke, Coil et al., in press

Keck/ESI: gas mass in each outflow

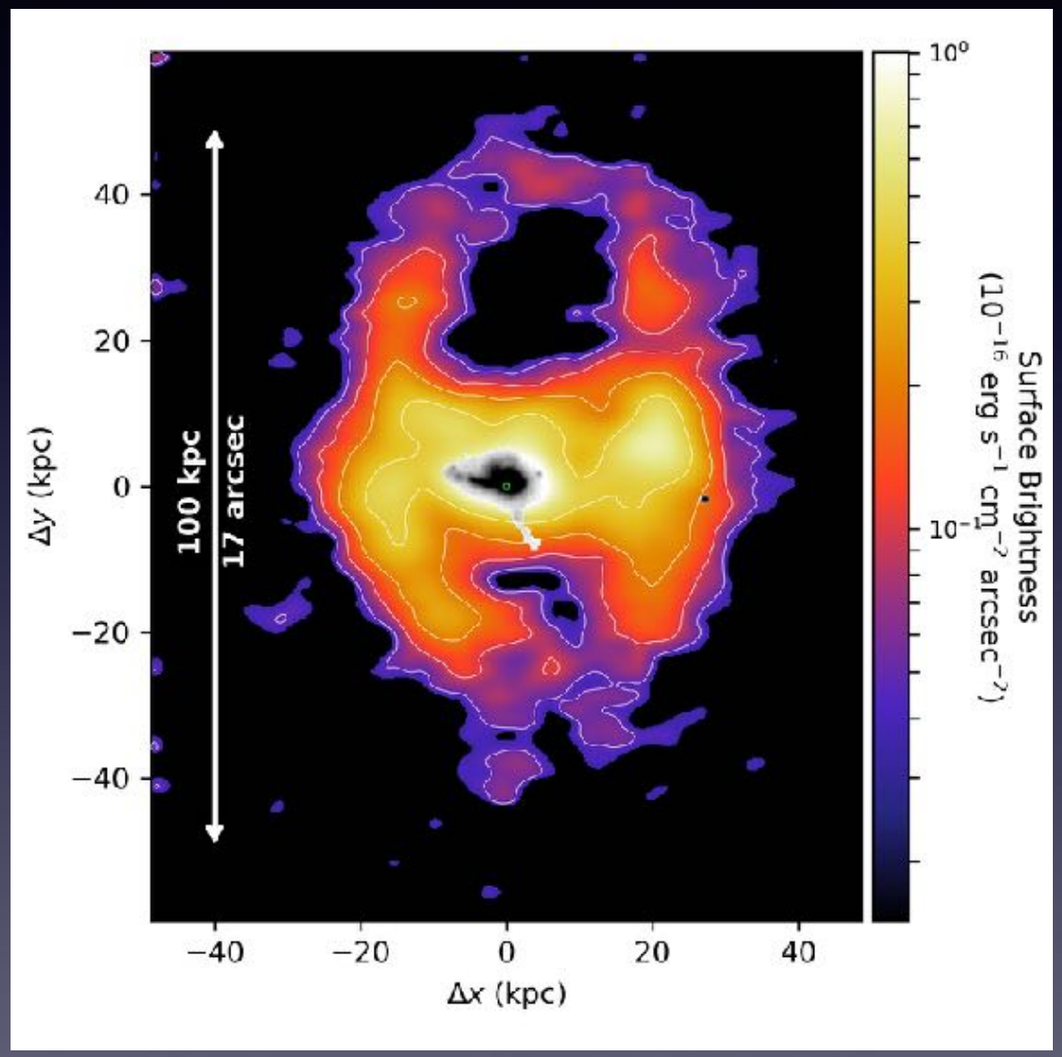
The James Webb Space Telescope: JWST



Webb's Specialization in Infrared Light



Observing Makani with JWST

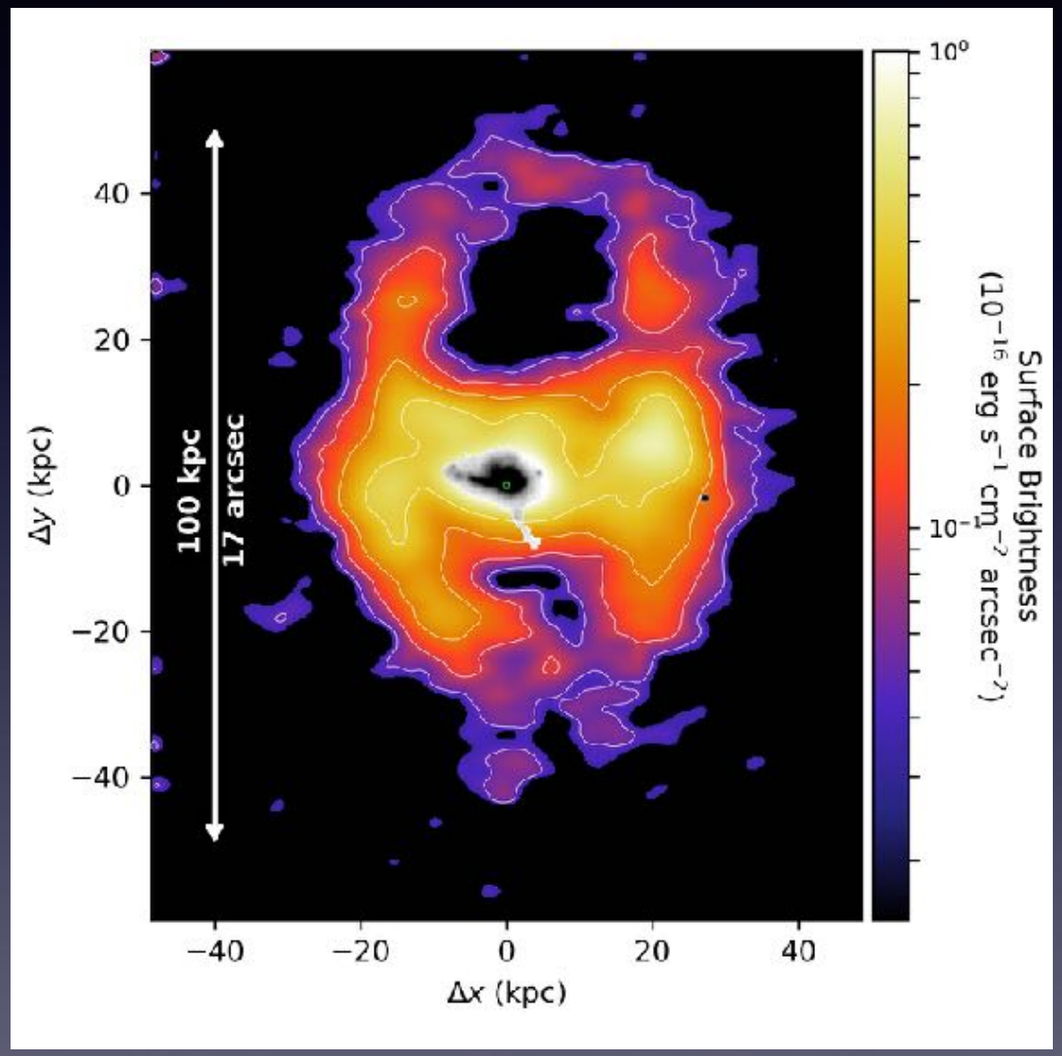


2 approved JWST proposals:

- MIRI/NIRCam imaging of the warm dust and cold gas that should be in the large outflow
- Would be first detection of dust around a galaxy on this scale, from an outflowing wind!

Help us understand what happens to the gas as it moves outward - dust is destroyed? Gas heats up?

Observing Makani with JWST



2nd approved JWST proposal:

- MIRI spectroscopy of 5 of our other galaxies to determine the drivers of the outflows

- Exploding stars? Supermassive black holes ejecting matter?

What is the star formation rate? What is the supermassive black hole accretion rate? What is driving these outflows?